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Neyhart

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(54) **NETWORK ACCESS COORDINATION OF
LOAD CONTROL DEVICES**

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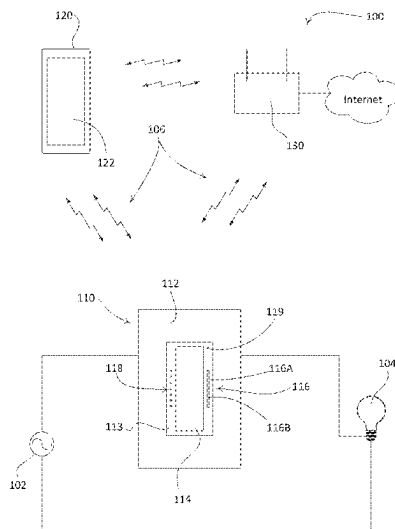
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(57) **ABSTRACT**

An apparatus may control the power delivered from an AC power source to an electrical load, and may comprise a controllably conductive device. The apparatus may also comprise a controller that may be operatively coupled to a control input of the controllably conductive device. The apparatus may also include a first wireless communication circuit operable to communicate via a first protocol and to join a first wireless communication network operable to communicate via the first protocol. The first wireless communication circuit may be in communication with the controller. The controller may be operative to determine a first condition for communicating via the first protocol. The controller may also be operable to control the first wireless communication circuit to join the first wireless communication network upon the first condition being satisfied.

30 Claims, 15 Drawing Sheets



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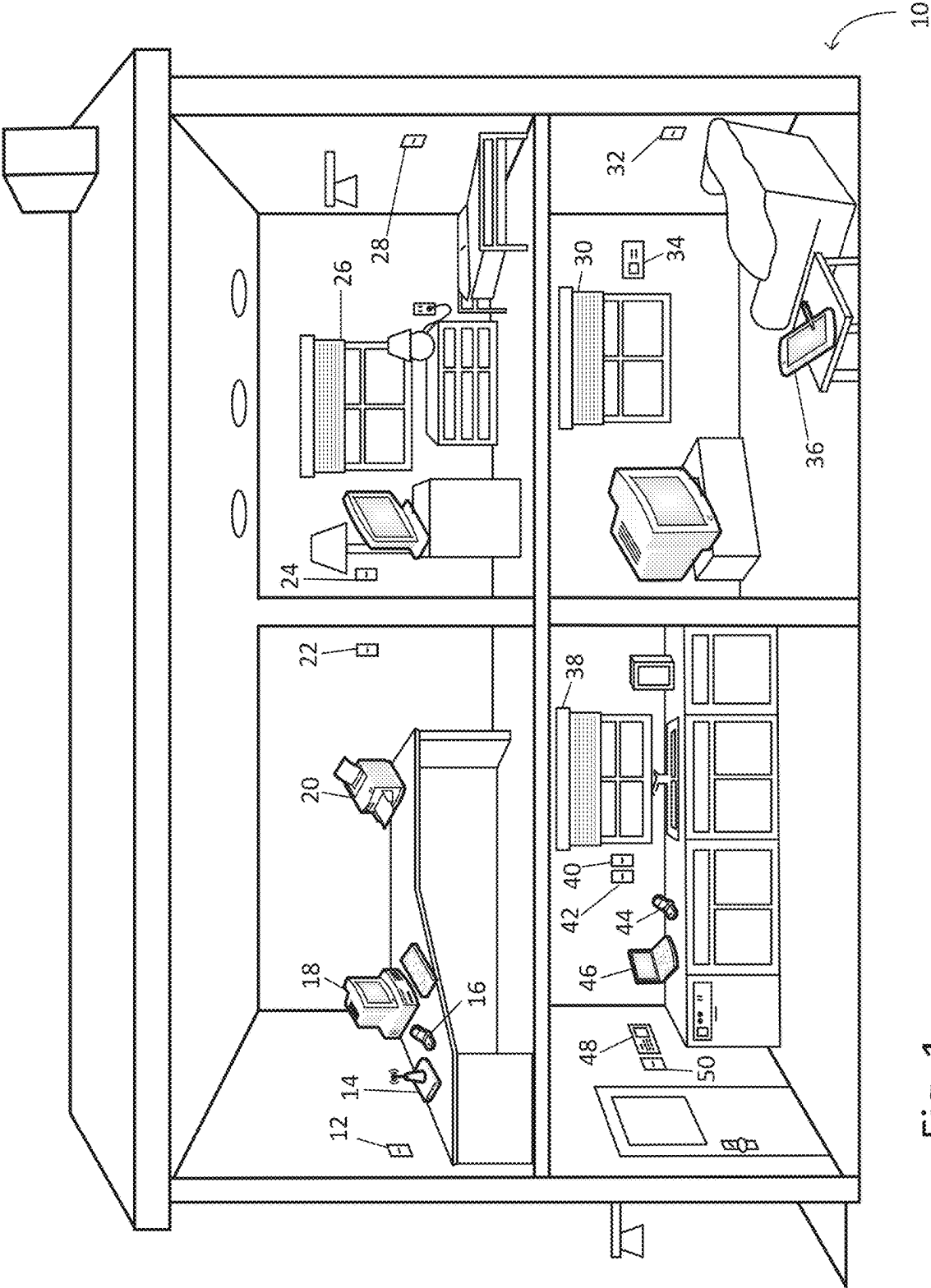


Fig. 1

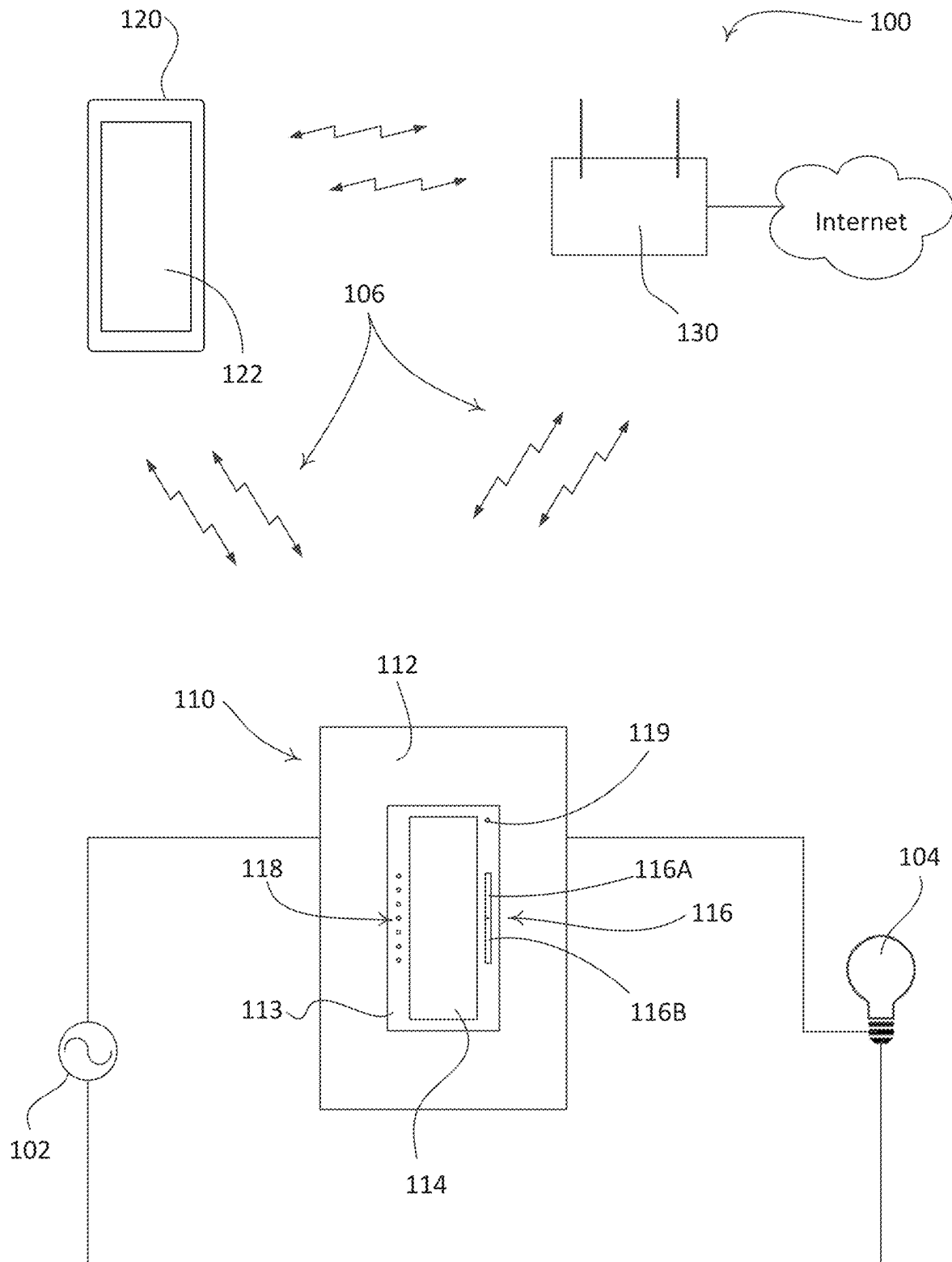


Fig 2

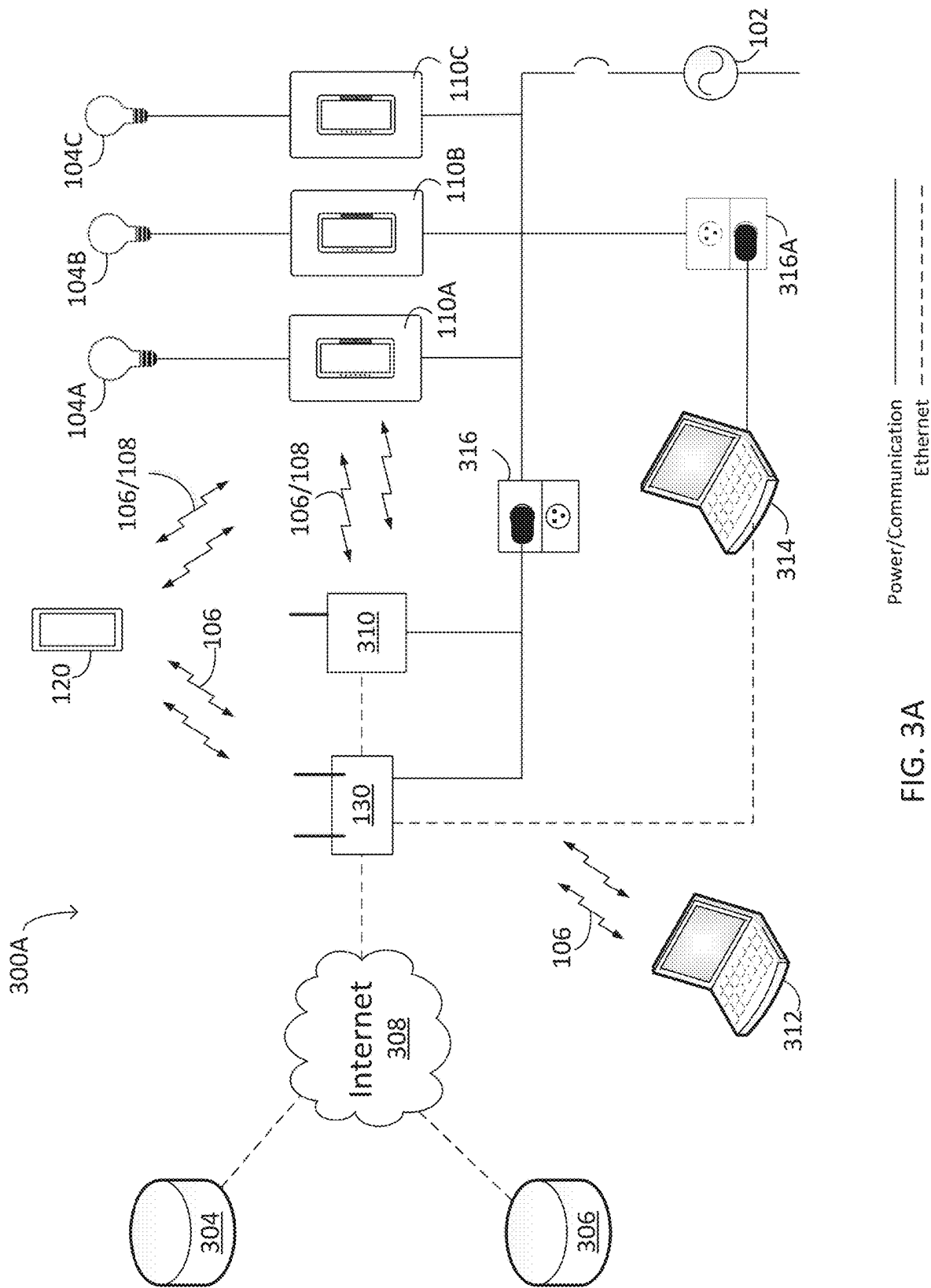
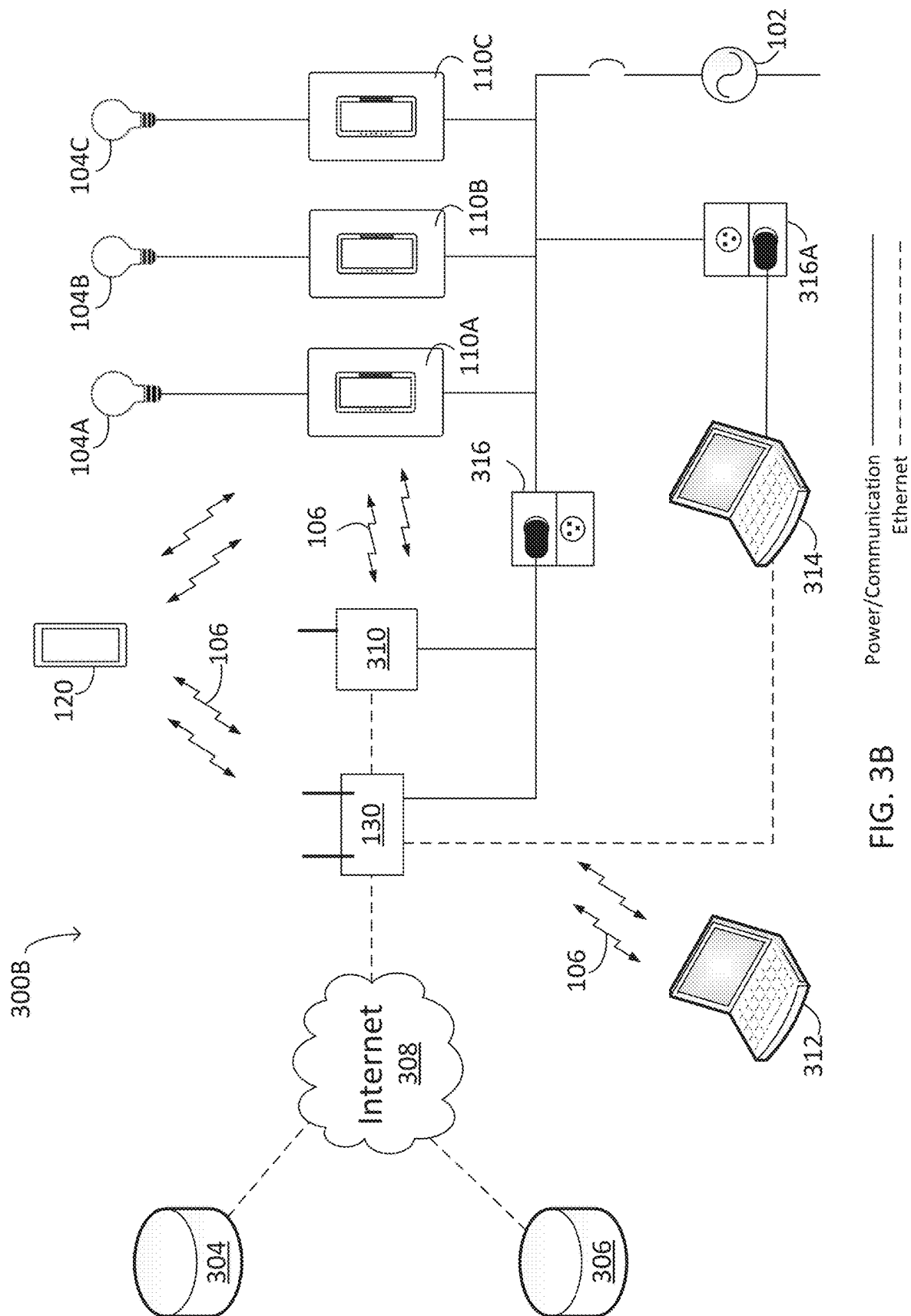
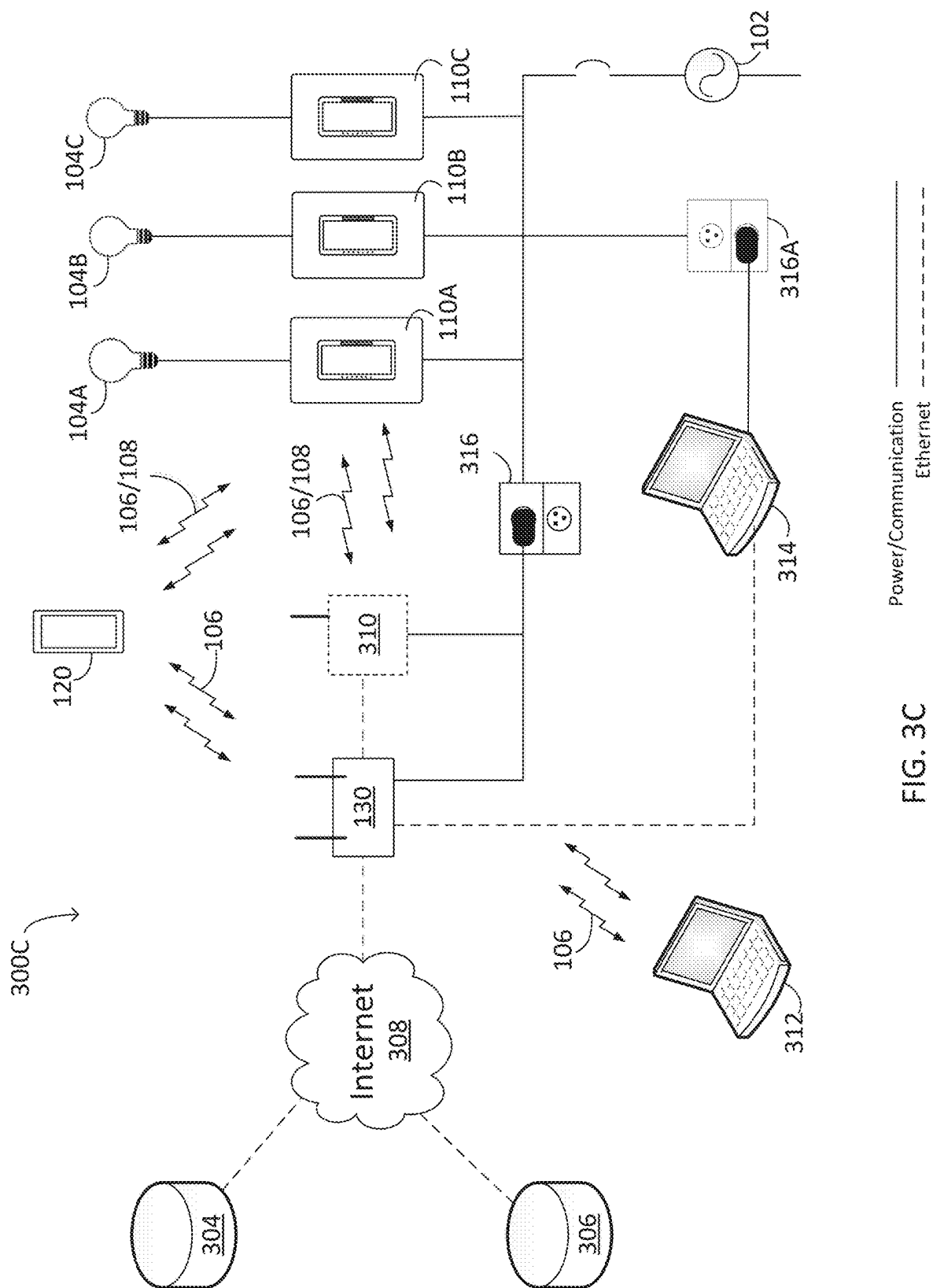


FIG. 3A





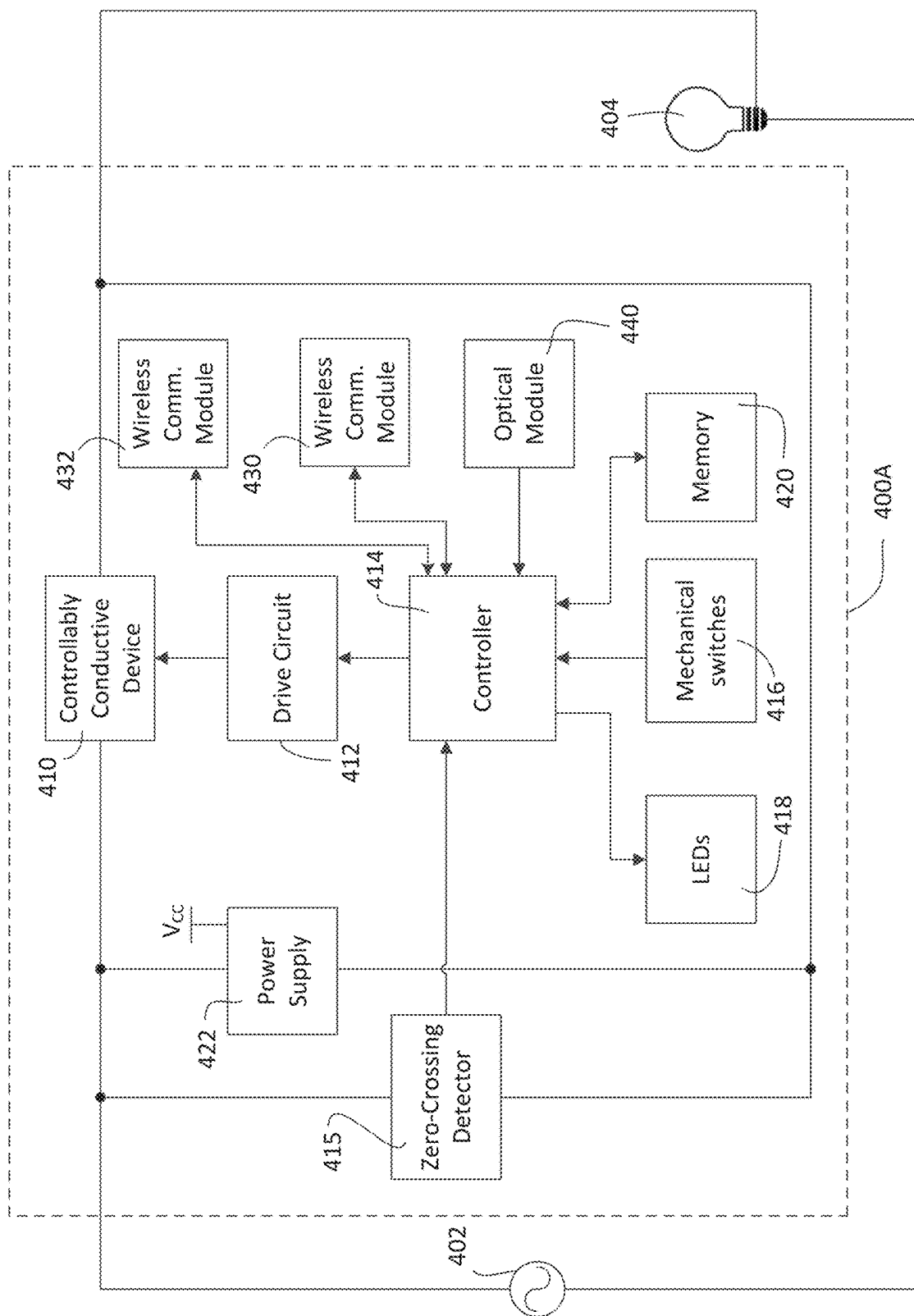


FIG. 4A

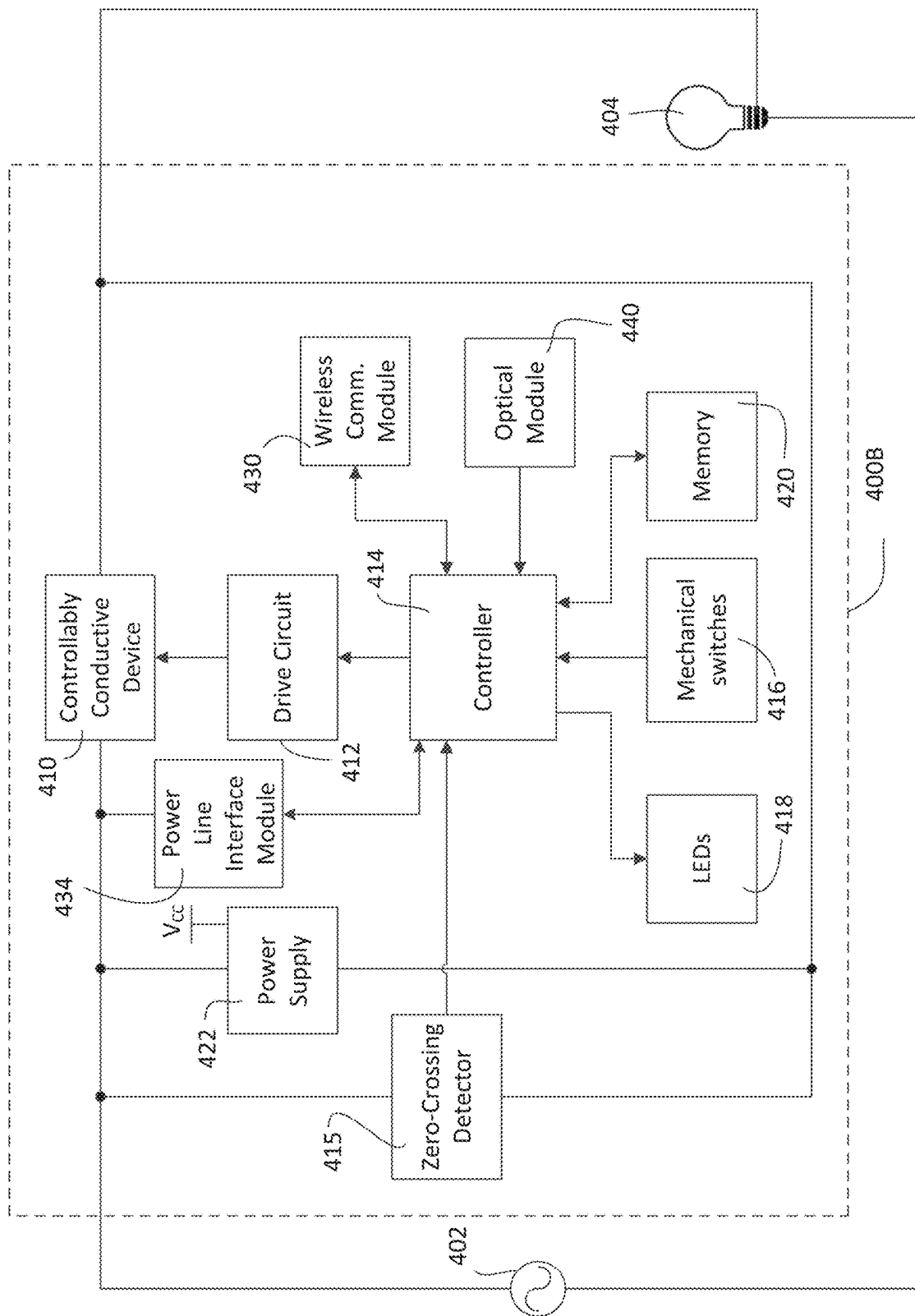


FIG. 4B

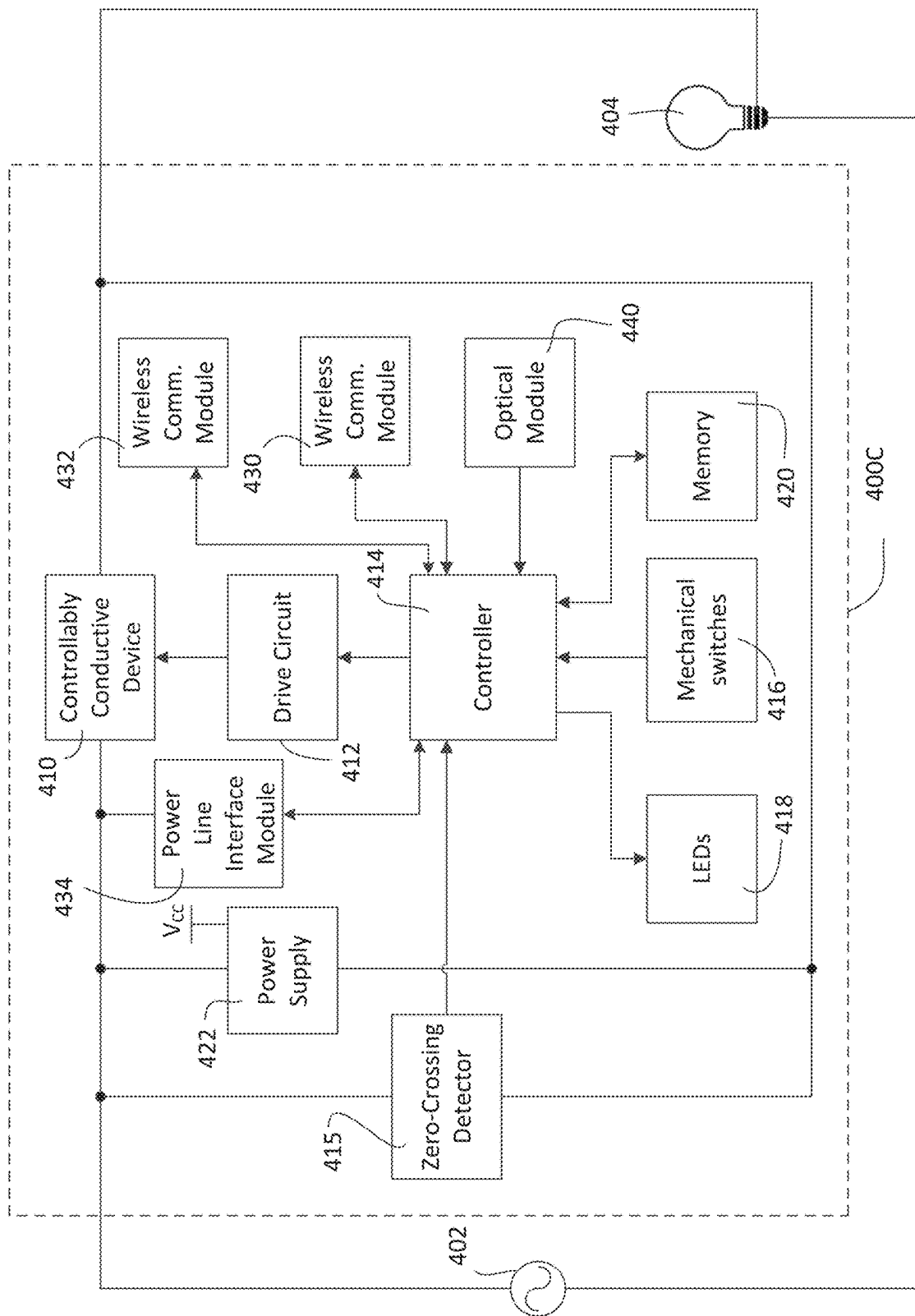


FIG. 4C

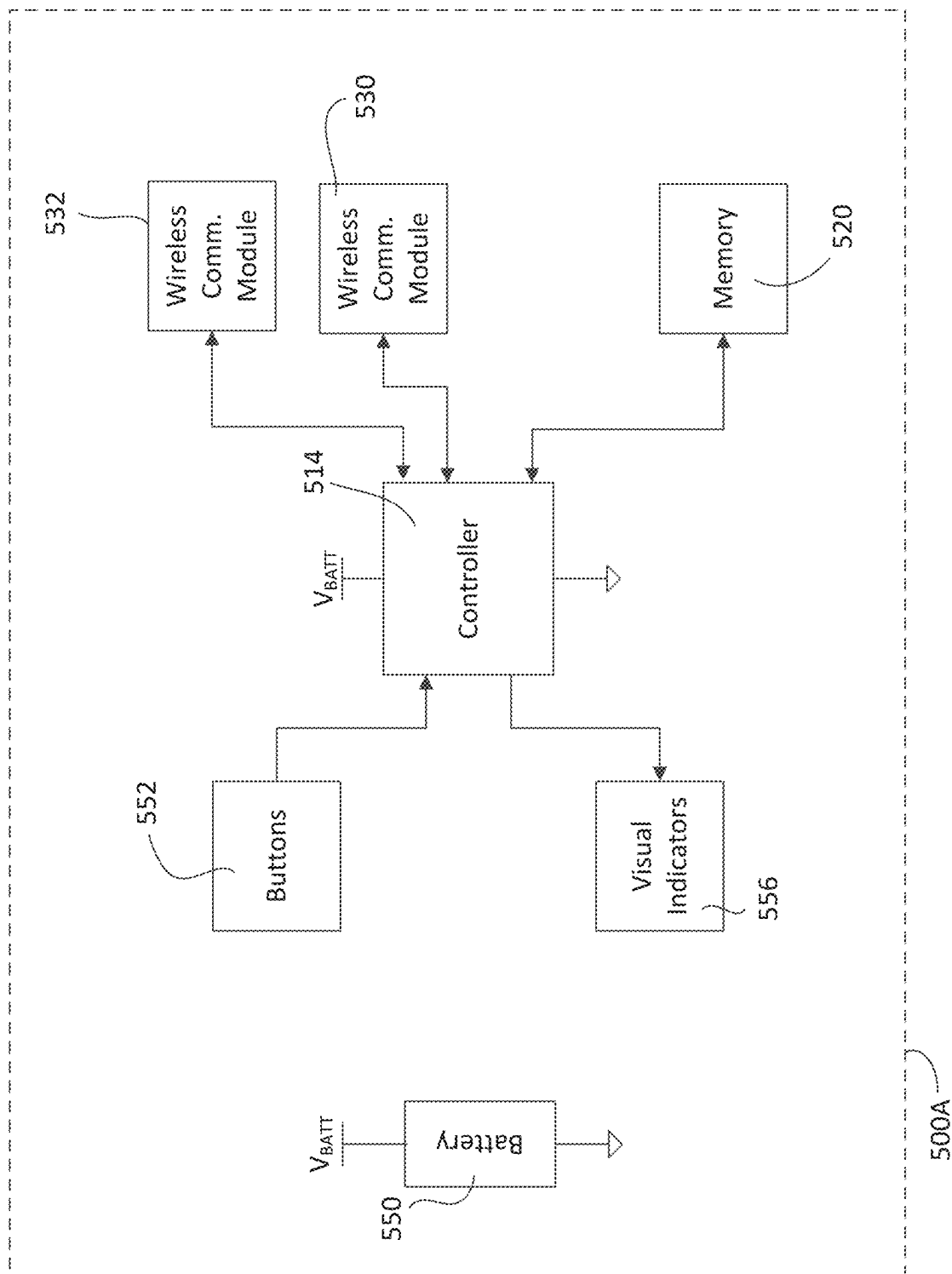


FIG. 5A

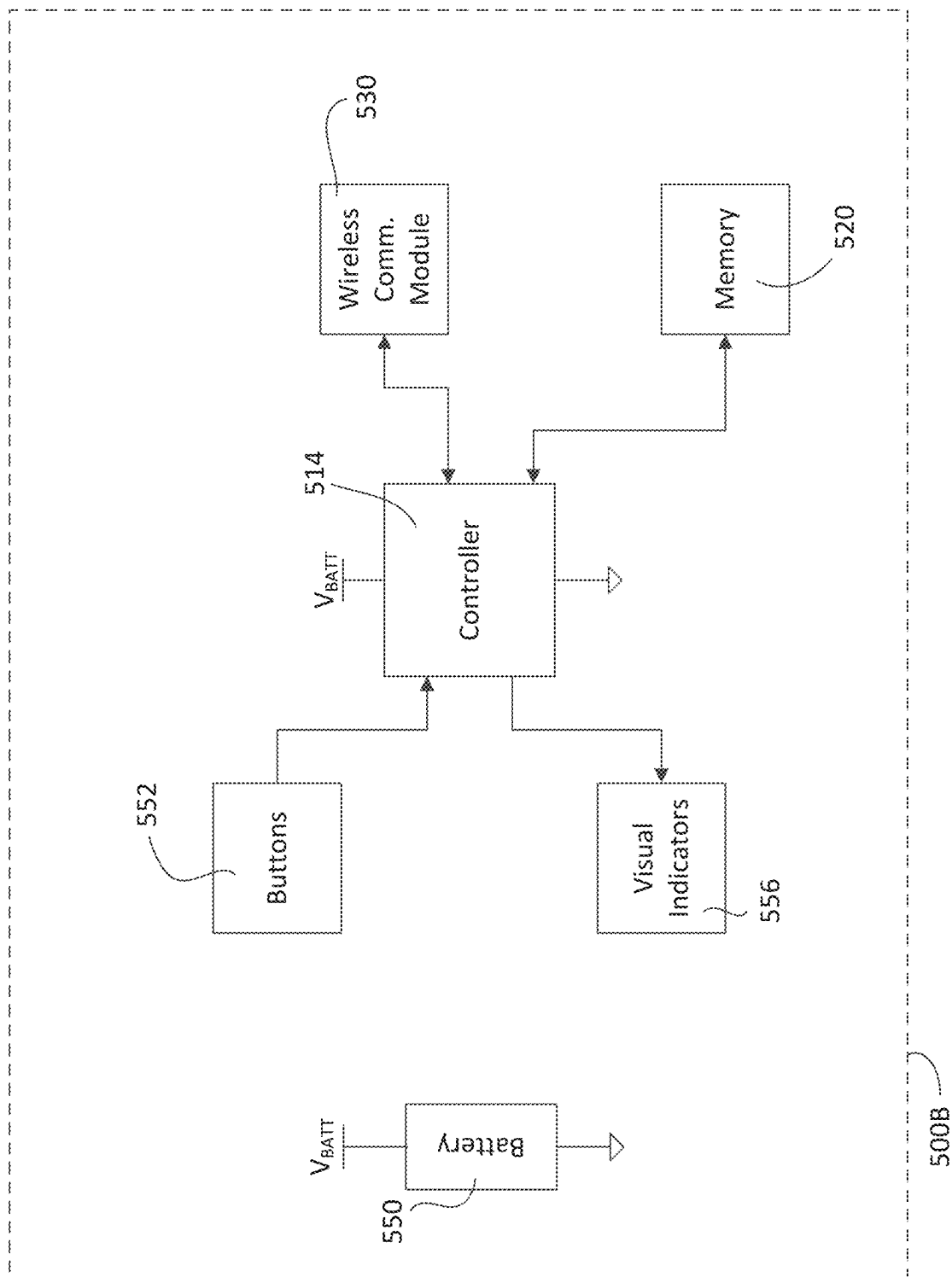


FIG. 5B

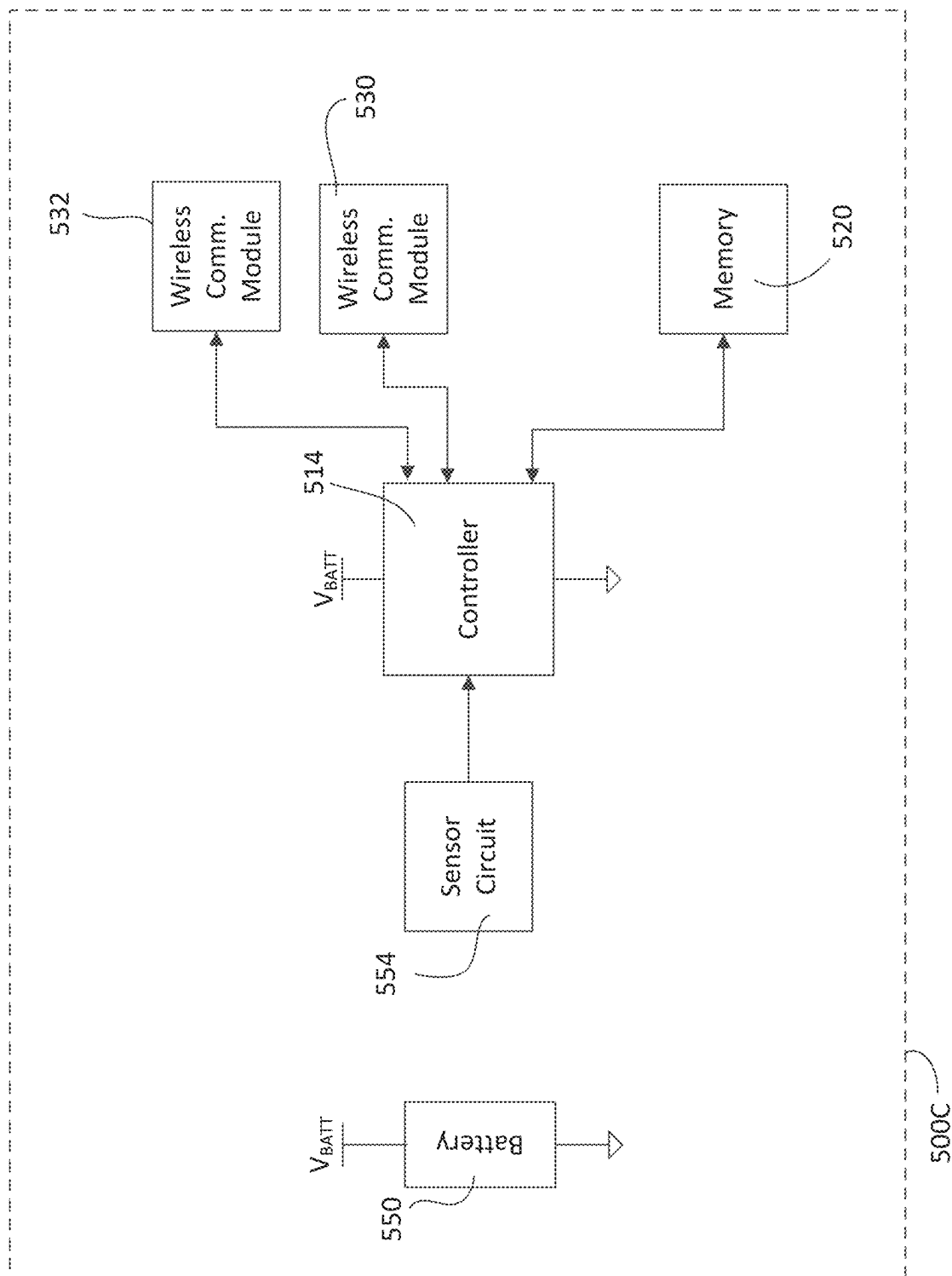


FIG. 5C

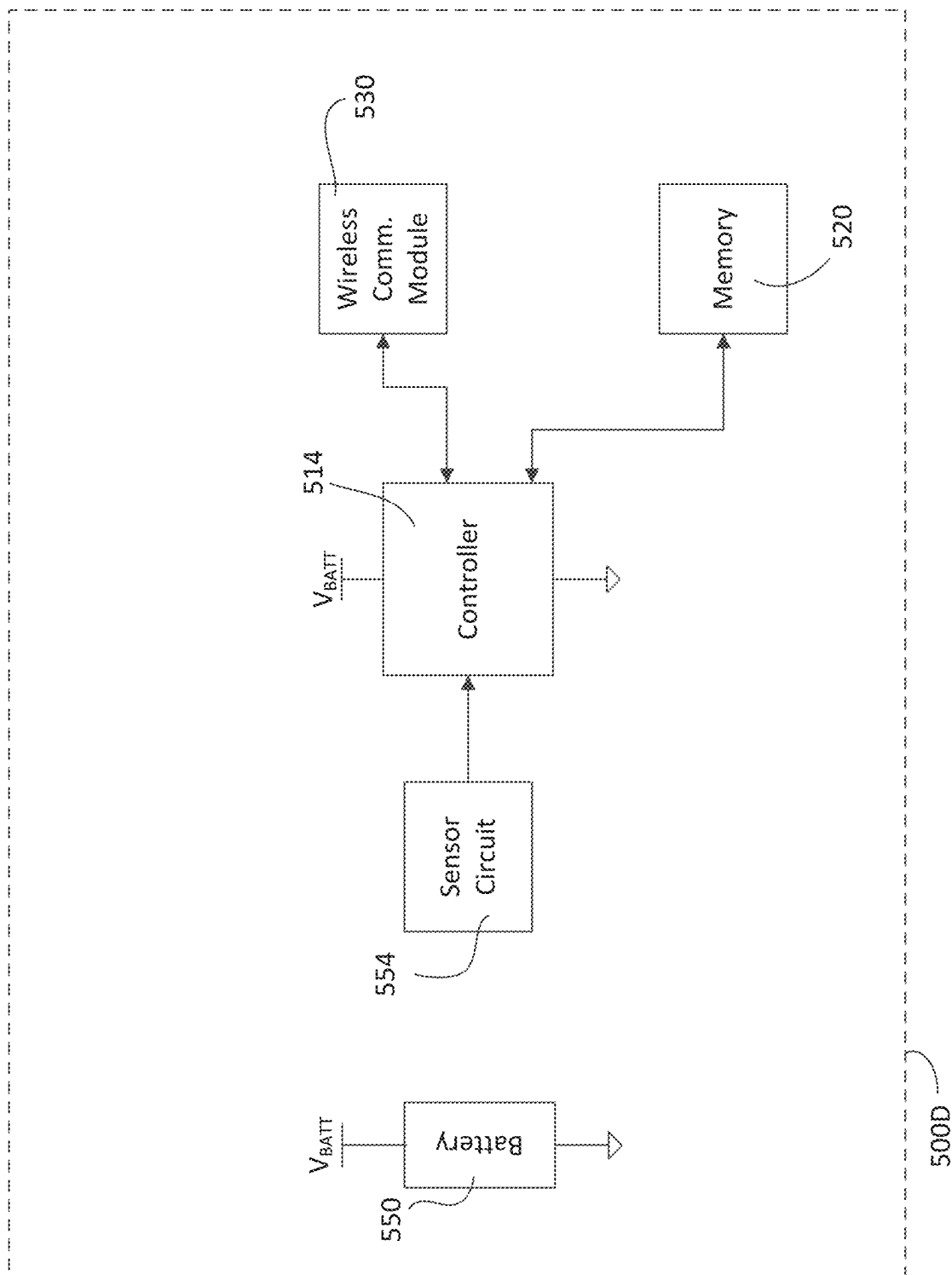


FIG. 5D

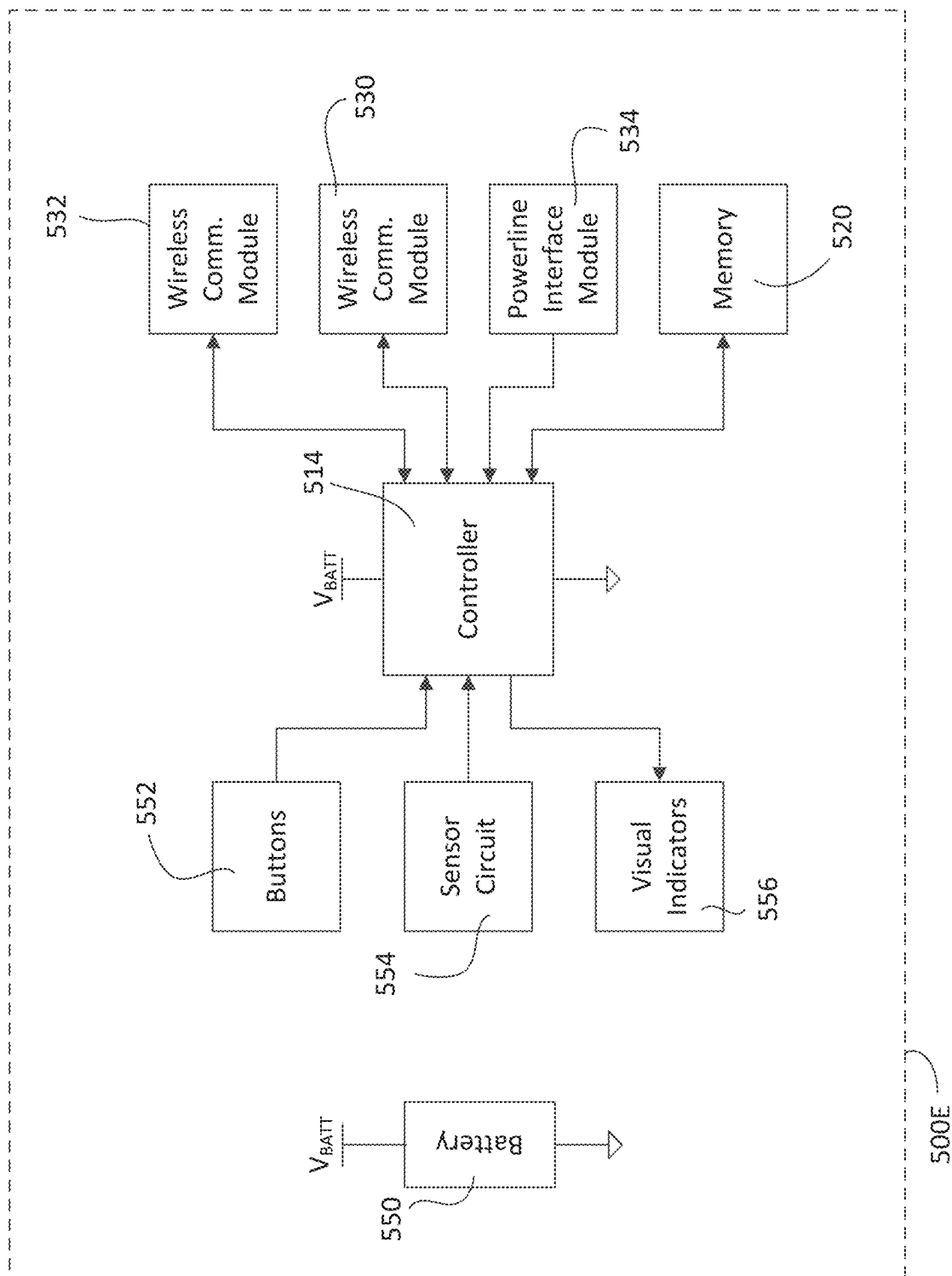


FIG. 5E

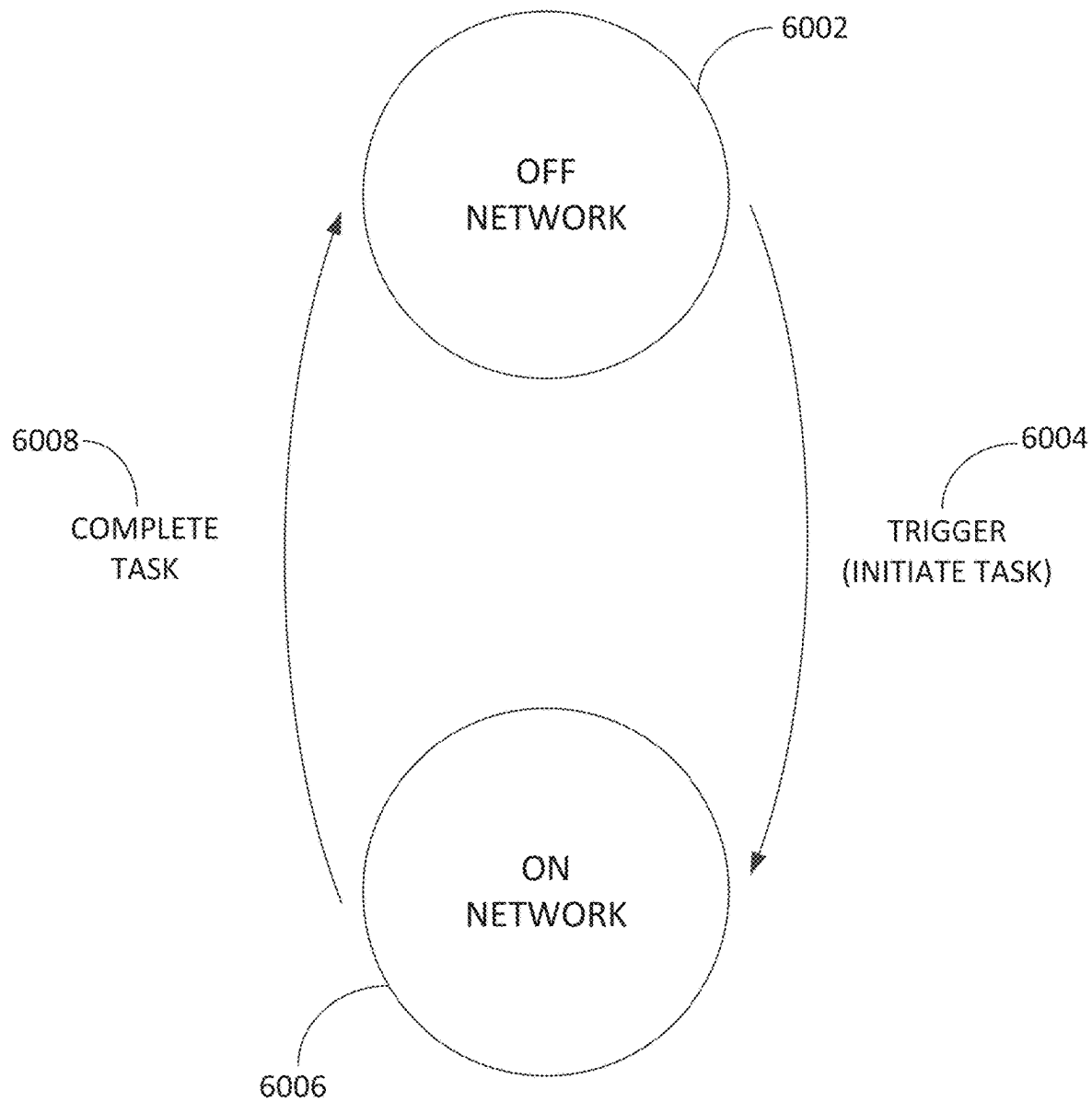


FIG. 6

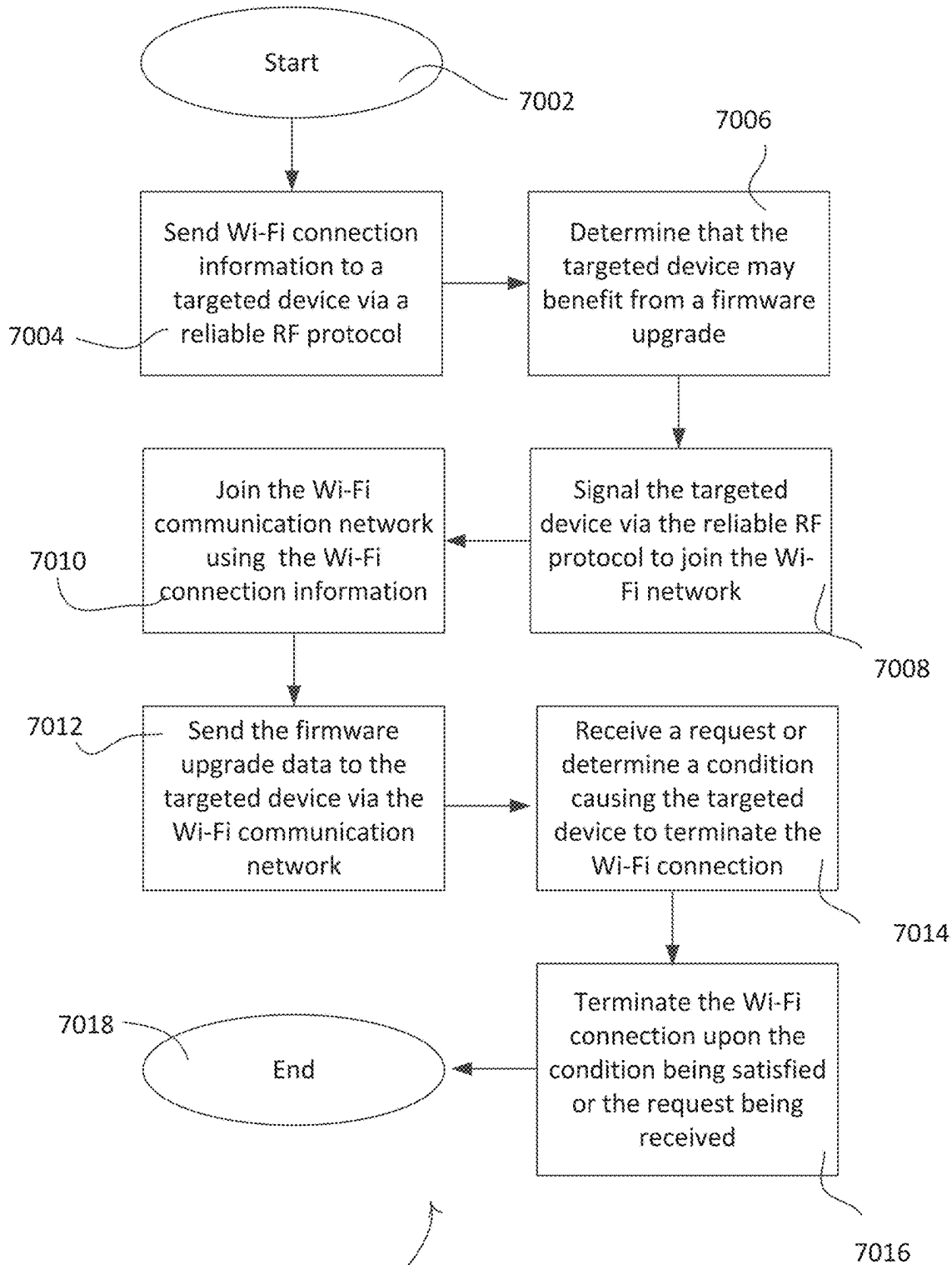


FIG. 7

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NETWORK ACCESS COORDINATION OF LOAD CONTROL DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. Non-Provisional application Ser. No. 16/989,145, filed on Aug. 10, 2020, which is a continuation of U.S. Non-Provisional application Ser. No. 16/102,357, filed on Aug. 13, 2018, (now U.S. Pat. No. 10,742,032, issued on Aug. 11, 2020), which is a continuation of U.S. Non-Provisional application Ser. No. 15/011,257 filed on Jan. 29, 2016, (now U.S. Pat. No. 10,050,444, issued Aug. 14, 2018), which is a continuation of U.S. Non-Provisional application Ser. No. 13/796,486 filed on Mar. 12, 2013 (now U.S. Pat. No. 9,413,171, issued Aug. 9, 2016), which claims the benefit of commonly assigned U.S. Provisional Patent Application No. 61/745,419, filed on Dec. 21, 2012, the entire contents of all of which being hereby incorporated by reference as if fully set-forth herein, for all purposes.

BACKGROUND

A load control device may control the amount of power delivered to an electrical load. Load control devices include, for example, lighting control devices (such as wall-mounted dimmer switches and plug-in lamp dimmers), motor control devices (for motor loads), temperature control devices, motorized window treatments, and remote controls. Typically, a load control device may be coupled in a series electrical connection between an alternating-current (AC) power source and the electrical load to control the power delivered from the AC power source to the electrical load.

In some applications, the load control device may connect to a wireless network, such as a Wi-Fi network for example. Examples of Wi-Fi-enabled load control devices include those described in commonly-assigned U.S. application Ser. No. 13/538,555, filed Jun. 29, 2012, titled LOAD CONTROL DEVICE HAVING INTERNET CONNECTIVITY, the contents of which is hereby incorporated by reference herein in its entirety, for all purposes. In practice, such devices may connect to wireless network provided by a wireless network access point (AP), such as an AP provided by a home Wi-Fi router.

APs, and particularly home Wi-Fi routers, often have practical capacity limitations well below the theoretical protocol maximums. For example, a typical IP subnet might theoretically support 254 addressable devices. However, a home Wi-Fi router may only have internal memory sized to support 30 addressable devices. These practical limitations are often unnoticed, even by the most voracious Internet households. Having more than 30 Wi-Fi devices, computers, tablets, cell phones, on a household network at one time is uncommon. And, even in a commercial setting, wireless networks are routinely engineered with 10-30 users per router. The traffic generated by 10-30 commercial users often reaches the practical traffic capacity for the router.

But, it is not uncommon for a residential or commercial installation to have well over 30 load control devices. FIG. 1 offers a partial illustration of the modern residential technological environment 10. In FIG. 1, traditional network devices such as computers 18 and 46, tablets 36, smart phones 16 and 44, and printer 20, when few in number may be well served by a home Wi-Fi router 14. However, a home using Wi-Fi-enabled load control devices such as lighting load controls 12, 22, 24, 28, 32, 40, 42, and 50, motorized

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window treatments 26, 30, and 38, smart thermostats 34 and 48, and the like, may have a total number of devices vying for network access from the home Wi-Fi router 14 that may well exceed the router's capacity.

SUMMARY

An apparatus, such as a dimmer switch, may control the power delivered from an AC power source to at least one electrical load, such as one or more lights. The apparatus may comprise a controllably conductive device adapted to be coupled in series electrical connection between the source and the one or more lights. The apparatus may also comprise a controller that may be operatively coupled to a control input of the controllably conductive device. The apparatus may also include a first wireless communication circuit that may communicate via a first protocol and may be used to join a first wireless communication network. The first wireless communication network may communicate via the first protocol. The first wireless communication circuit may be in communication with the controller. The controller may control the controllably conductive device for rendering the controllably conductive device conductive and non-conductive, perhaps to increase and/or decrease the intensity of the one or more lights. The controller may also determine a first condition for communicating via the first protocol. The controller may control the first wireless communication circuit to join the first wireless communication network when the first condition being is.

An apparatus, such as a remote control device or an occupancy sensor, may be configured to provide information for the control of power delivered to at least one electrical load, such as sending commands to dimmer switches. The apparatus may comprise a controller and a sensor (e.g. for the occupancy sensor) or a manual operator (e.g., for the remote control device). Either the sensor and/or the manual operator may be in communication with the controller. The apparatus may further comprise a first wireless communication circuit that may communicate via a first protocol and may join a first wireless communication network. The first wireless communication network may be operable for communication via the first protocol. And the first wireless communication circuit may communicate with the controller. The controller may determine the information based at least in part on a signal received from either the sensor and/or the manual operator. The controller may also determine a first condition for communicating via the first protocol. And the controller may also control the first wireless communication circuit to join the first wireless communication network when the first condition is satisfied.

An apparatus may control the power delivered to at least one electrical load. The apparatus may comprise a controller and a first wireless communication circuit that may be operable to communicate via a first protocol and to join a first wireless communication network that may be operable to communicate via the first protocol. The first communication circuit may be in communication with the controller. The apparatus may also comprise a second communication circuit operable to communicate via a second protocol. The second communication circuit may be in communication with the controller. The controller may be operable to determine a first condition for communicating via the first protocol. The first condition may include a receipt of a signal via the second communication circuit and via the second protocol to join the first wireless communication network. The controller may also be operable to control the first

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wireless communication circuit to join the first wireless communication network upon the first condition being satisfied.

A network node may be in communication with a load control device. The load control device may control the power delivered to at least one electrical load. The load control device may comprise a first controller. The load control device may also comprise a first wireless communication circuit that may be operable to communicate via first protocol and may be operable to join a first wireless communication network. The load control device may also comprise a second communication circuit that may be operable to communicate via a second protocol. The network node may comprise a second controller and a third communication circuit that may be operable to communicate via the second protocol. The third communication circuit may be in communication with the second controller. The second controller may be operable to determine a first condition for the load control device to communicate via the first protocol. The second controller may also be operable to send a first signal via the third communication circuit and via the second protocol to the load control device upon the first condition being satisfied. The first signal may cause the load control device to control the first wireless communication circuit to join the first wireless communication network.

A wireless control device may be used in a load control system to control power delivered from a power source to an electrical load. The wireless control device may comprise a first wireless communication circuit that may be configured to communicate digital messages via a first wireless communication network. The wireless device may comprise a control circuit that may be in communication with the first wireless communication circuit. The control circuit may be configured to join the first wireless communication network. The control circuit may be configured to communicate digital messages via the first wireless communication network using a first protocol. The control circuit may be configured to determine an occurrence of a first condition. The control circuit may be configured to join the first wireless communication network upon the first condition being satisfied. The control circuit may be configured to determine an occurrence of a second condition. The control circuit may be configured to disconnect from the first wireless communication network upon the second condition being satisfied.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an example environment that may utilize a number of contemplated load control devices, sensors, and/or remote control devices.

FIG. 2 is a simple diagram of a radio-frequency (RF) lighting control system comprising a dimmer switch and a wireless control device, such as a smart phone.

FIG. 3A is a diagram of a first example network in which one or more contemplated devices and techniques may be employed.

FIG. 3B is a diagram of a second example network in which one or more contemplated devices and techniques may be employed.

FIG. 3C is a diagram of a third example network in which one or more contemplated devices and techniques may be employed.

FIG. 4A is a first simplified example block diagram of the dimmer switch of the RF lighting control system of FIG. 2.

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FIG. 4B is a second simplified example block diagram of the dimmer switch of the RF lighting control system of FIG. 2.

FIG. 4C is a third simplified example block diagram of the dimmer switch of the RF lighting control system of FIG. 2.

FIG. 5A is a first simplified example block diagram of an input device like the remote control devices of FIGS. 3A-3C.

FIG. 5B is a second simplified example block diagram of an input device like the remote control device of FIGS. 3A-3C.

FIG. 5C is a first simplified example block diagram of a sensor device like the occupancy sensor of FIGS. 3A-3C.

FIG. 5D is a second simplified example block diagram of a sensor device like the occupancy sensor of FIGS. 3A-3C.

FIG. 5E is a simplified example block diagram of a contemplated combination input and sensor device which may be employed in the environments of FIGS. 3A-3C.

FIG. 6 is a diagram that illustrates network access activity for one or more contemplated devices and techniques.

FIG. 7 is an exemplary flow chart for Internet Protocol address assignments for load control devices.

DETAILED DESCRIPTION

FIG. 2 is a simple diagram of a radio-frequency (RF) lighting control system 100 that may include a dimmer switch 110 and a wireless control device 120. The wireless control device 120 may be any device capable of performing wireless communications, such as, a smart phone (e.g., an iPhone® smart phone, an Android® smart phone, or a BlackBerry® smart phone), a personal computer, a laptop, a wireless-capable media device (e.g., MP3 player, gaming device, or television), or a tablet device, (for example, an iPad® hand-held computing device), a Wi-Fi or wireless-communication-capable television, or any other suitable Internet-Protocol-enabled device.

The wireless control device 120 may be operable to transmit digital messages to the dimmer switch 110 in one or more Internet Protocol (IP) packets. The Internet Protocol layer is responsible for addressing hosts and routing datagrams (i.e., packets) from a source host to a destination host across one or more IP networks. For this purpose, the Internet Protocol layer defines an addressing system that has two functions: identifying hosts and providing a logical location service. This is accomplished by defining standard datagrams and a standard addressing system.

Each datagram has two components, a header and a payload. The IP header includes the source IP address, destination IP address, and other meta-data needed to route and deliver the datagram. The payload is the data that is transported.

The wireless control device 120 may transmit the digital messages (e.g., IP packets) via RF signals 106 either directly or via a wireless network that includes a standard wireless router 130. For example, the wireless control device 120 may transmit the RF signals 106 directly to the dimmer switch 110 via a point-to-point communication link, e.g., a Wi-Fi communication link, such as an 802.11 wireless local area network (LAN), or other direct wireless communication link, such as a Wi-MAX communication link or a Bluetooth® communication link. This point-to-point communication may be performed using a standardized communication, e.g., Wi-Fi Direct communication, or any non-standardized communication that allows a wireless device to connect to another wireless device without the use of a wireless access point. For example, the wireless control

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device **120** and/or the dimmer switch **110** may download a software access point (AP) that provides a protected wireless communication between the devices.

The wireless control device **120** may also transmit RF signals **106** to the dimmer switch **110** via the wireless network (i.e., via the wireless router **130**). The wireless network may enable wireless communications via one or more wireless communications links, e.g., a Wi-Fi communications link, a Wi-MAX communications link, a Bluetooth® communications link, a cellular communications link, a television white space (TVWS) communication link, or any combination thereof. For example, the wireless control device **120** may communicate with a network server via a first wireless communications link (e.g., a cellular communications link), while the dimmer switch **110** communicates with the network server via a second communications link (e.g., a Wi-Fi communications link). Alternatively or additionally, the wireless control device **120** and the dimmer switch **110** may communicate with the network via the same type of communication link. The lighting control system **100** may also include a femtocell, a Home Node B, and/or other network entity for facilitating the configuration and operation of the lighting control system and for allowing wireless communications and connection to the Internet.

The dimmer switch **110** may be coupled in series electrical connection between an AC power source **102** and a lighting load **104** for controlling the amount of power delivered to the lighting load. The dimmer switch **110** may be wall-mounted in a standard electrical wallbox, or alternatively implemented as a table-top load control device. The dimmer switch **110** comprises a faceplate **112** and a bezel **113** received in an opening of the faceplate. The dimmer switch **110** further comprises a toggle actuator **114** and an intensity adjustment actuator **116**. Actuations of the toggle actuator **114** toggle, e.g., alternately turn off and on, the lighting load **104**. Actuations of an upper portion **116A** or a lower portion **116B** of the intensity adjustment actuator **116** may respectively increase or decrease the amount of power delivered to the lighting load **104** and thus increase or decrease the intensity of the lighting load **104** from a minimum (i.e., low-end) intensity (e.g., approximately 1-10%) to a maximum (i.e., high-end) intensity (e.g., approximately 100%). A plurality of visual indicators **118**, e.g., light-emitting diodes (LEDs), may be arranged in a linear array on the left side of the bezel **113**. The visual indicators **118** are illuminated to provide visual feedback of the intensity of the lighting load **104**. An example of a dimmer switch having a toggle actuator and an intensity adjustment actuator is described in greater detail in U.S. Pat. No. 5,248,919 ("the 919 patent"), issued Sep. 28, 1993, entitled LIGHTING CONTROL DEVICE, the entire disclosure of which is hereby incorporated by reference. Alternatively, the dimmer switch **110** could be replaced by an electronic switch for simply turning the lighting load **104** on and off. The electronic switch may include a single visual indicator, e.g., the middle indicator of the visual indicators **118** of the dimmer switch **110**.

The dimmer switch **110** may include an optical receiver **119**. The optical receiver **119** may be used to receive optical signals from the wireless control device **120**. Optical signals may be free-space optical communications or communications via physical connections. For example, free space optical communications may include communications via air, while physical optical communications may include communications via optical fiber cable or an optical trans-

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mission pipe. The optical signals may also be included in visible light, e.g., a flashing light, or non-visible light, e.g., infrared, spectrums.

The optical signals may provide instructions for programming and/or adjusting the operating parameters (e.g., the low-end intensity and the high-end intensity) of the dimmer switch **110**. For example, the optical signals may be used to configure the dimmer switch such that the dimmer switch **110** is operable to receive the RF signals **106** from the wireless control device **120** as will be described in greater detail below. The optical signals may also be used to control or program the lighting configurations of the dimmer switch **110**. And, though devices and techniques described herein may be described with respect to using optical signals or other signals to program or control a dimmer switch from a wireless control device, such signals may be used to program or control any device that is capable of receiving instructions via such optical or other signals, such as shades, thermostats, plug-in devices, or the like. Examples of methods of communicating optical signals between the dimmer switch **110** and the wireless control device **120** are described in greater detail in commonly assigned U.S. patent application Ser. No. 13/538,665, filed on Jun. 29, 2012, titled METHOD OF OPTICALLY TRANSMITTING DIGITAL INFORMATION FROM A SMART PHONE TO A CONTROL DEVICE, the entire disclosure of which is hereby incorporated by reference.

Wireless load control devices are described in greater detail in commonly-assigned U.S. Pat. No. 5,838,226, issued Nov. 17, 1998, entitled COMMUNICATION PROTOCOL FOR TRANSMISSION SYSTEM FOR CONTROLLING AND DETERMINING THE STATUS OF ELECTRICAL DEVICES FROM REMOTE LOCATIONS; U.S. Pat. No. 6,803,728, issued Oct. 12, 2004, entitled SYSTEM FOR CONTROL OF DEVICES; U.S. patent application Ser. No. 12/033,223, filed Feb. 19, 2008, entitled COMMUNICATION PROTOCOL FOR A RADIO-FREQUENCY LOAD CONTROL SYSTEM; and U.S. patent application Ser. No. 13/234,573, filed Sep. 16, 2011, entitled DYNAMIC KEY-PAD FOR CONTROLLING ENERGY-SAVINGS SETTINGS OF A LOAD CONTROL SYSTEM; the entire disclosures of which are hereby incorporated by reference.

The wireless control device **120** has a visual display **122**, which may comprise a touch screen having, for example, a capacitive touch pad displaced overtop the visual display, such that the visual display may display soft buttons that may be actuated by a user. Alternatively, the wireless control device **120** may comprise a plurality of hard buttons (e.g., physical buttons or manual operators) in addition to the visual display **122**. The wireless control device **120** may download a product control application for allowing the user to control the lighting load **104**. In response to actuations of the displayed soft buttons or hard buttons, the wireless control device **120** transmits digital messages to the dimmer switch **110** directly or through other wireless communications described herein. For example, the digital messages may be transmitted via Wi-Fi communication using the wireless router **130**. The dimmer switch **110** may adjust the intensity of the lighting load **104** in response to commands included in the digital messages, such that the dimmer switch controls the lighting load in response to actuations of the soft buttons or hard buttons of the wireless control device **120**.

In addition, the wireless control device **120** may be controlled to transmit optical signals, near field communication (NFC) signals, or RF signals according to a proprietary RF communication protocol (such as, for example, the

Clear Connect™ protocol) as described herein. For example, the visual display **122** may be controlled to transmit optical signals to the optical receiver **119** of the dimmer switch **110** (as will be described in greater detail below).

The dimmer switch **110** and the wireless control device **120** may both be assigned a unique address for wireless communications via the wireless network (i.e., via the wireless router **130**) as described herein. For example, where wireless communications are performed using a Wi-Fi communication link, a Media Access Control (MAC) address may be assigned (e.g., during manufacture). The wireless control device **120** may connect to the wireless LAN via the wireless router **130** using standard procedures. The wireless control device **120** is assigned an Internet Protocol (IP) address upon connecting to the wireless LAN. The wireless control device **120** may store the service set identifier (SSID) and the SSID password of the wireless LAN. After obtaining the IP address, the wireless control device **120** is able to assign an IP address (e.g., different from the IP address of the wireless control device **120**) to the dimmer switch **110**. Alternatively, the dimmer switch **110** may be operable to obtain the IP address from the wireless router **130** using, for example, procedures defined by the Wi-Fi Protected Setup standard.

The dimmer switch **110** may be associated with (e.g., assigned to) the wireless control device **120**, such that the wireless control device may transmit commands for controlling the intensity of the lighting load **104** or programming the dimmer switch **110**. Such commands may be transmitted to the dimmer switch **110** via the RF signals **106**. Digital messages transmitted to and from the dimmer switch **110** may include, for example, the MAC address and the IP address of the dimmer switch **110**. The dimmer switch **110** is operable to turn the lighting load **104** on and off. The dimmer switch **110** is also operable to adjust the intensity of the lighting load in response to received digital messages, including the MAC address and the IP address of the dimmer switch, for example. In addition, the wireless router **130** may be operable to receive commands for controlling the lighting load **104** from the Internet, and may wirelessly transmit corresponding digital messages to the dimmer switch **110**.

The dimmer switch **110** may be assigned an IP address, an SSID, an SSID password, and/or a software AP at manufacture, such that the dimmer switch **110** may act as an AP for other communication devices in a LAN. The wireless control device **120** may recognize the dimmer switch **110** as an AP and may connect to the LAN via the dimmer switch **110**. For example, the dimmer switch **110** may connect to router **130** or may perform the functions of the router **130** itself.

The dimmer switch **110** may also connect to the wireless LAN to discover other dimmer switches (not shown). The dimmer switch **110** may discover the other dimmer switches using any discovery protocol, such as Bonjour, Simple Service Discovery Protocol (SSDP), Bluetooth® Service Discovery Protocol (SDP), DNS service discovery (DNS-SD), Dynamic Host Configuration Protocol (DHCP), Internet Storage Name Service (iSNS), Jini for Java objects, Service Location Protocol (SLP), Session Announcement Protocol (SAP) for RTP sessions, Simple Service Discovery Protocol (SSDP) for Universal Plug and Play (UPnP), Universal Description Discovery and Integration (UDDI) for web services, Web Proxy Autodiscovery protocol (WPAD), Web Services Dynamic Discovery (WS-Discovery), XMPP Service Discovery (XEP-0030), and/or XRDS for XRI, OpenID, OAuth, etc. Upon the dimmer switch **110** discovering one or more other dimmer switches, the dimmer switch

may create a peer-to-peer network of dimmer switches capable of communicating with one another. For example, the dimmer switches may communicate programming and/or control instructions received from the wireless control device **120**.

The wireless control device **120** may control the lighting load **104** by communicating instructions to the dimmer switch **110** via the RF signals **106** that cause the dimmer switch **110** to execute control instructions that have been pre-programmed on the dimmer switch **110**. For example, the dimmer switch **110** may be pre-programmed at manufacture or via an update to execute the control instructions. The control instructions may include pre-configured settings (e.g., protected or locked lighting presets), instructions for raising/lowering lighting level, instructions for fading, instructions for scheduling, instructions for turning lights on/off, or any other pre-programmed instruction, for example.

The wireless control device **120** may also program the settings (i.e., the operating parameters) of the dimmer switch **110** (e.g., when the dimmer switch is in a programming mode). For example, the dimmer switch **110** may be a dimmer switch that may have a limited user interface (UI) or may not have any user interface. As such, the user interface of the wireless control device **120** may be used to program the dimmer switch **110**. For example, various wireless communication links described herein, e.g., Wi-Fi signals, optical signals, near field communication (NFC) signals, or proprietary-protocol RF signals, may be used to program any of a number of programmable features provided by the dimmer switch **110**. Such features may be selected via the wireless control device **120**. For example, the wireless control device **120** may program the dimmer switch **110** with such features as protected or locked presets, high-end trim, low-end trim, adjustable delay, fade time, load type, performing communications via wireless communication modes (e.g., as described herein), or being compatible with different lamps. In addition, the wireless control device **120** may be operable to program the dimmer switch **110** to change between modes of operation, for example, between a switching mode, a dimming mode, and/or an electronic timer mode (i.e., a countdown timer mode). The programming signal may be a one-way or two-way serial communication with the dimmer switch **110**. Examples of method of programming the dimmer switch **110** using the wireless control device **120** are described in greater detail in commonly assigned U.S. patent application Ser. No. 13/538,615, filed Jun. 29, 2012, titled METHOD OF PROGRAMMING A LOAD CONTROL DEVICE USING A SMART PHONE, the entire disclosure of which is hereby incorporated by reference.

FIG. 3A is a diagram of an exemplary network environment **300A**. In FIG. 3A, the router **130** may communicate with one or more servers **304**, **306** via the Internet **308**, perhaps as accessed through the “cloud.” For example, router **130** may establish at least one Internet Protocol respective public IP addresses of server **304** and/or server **306**. In some configurations, a gateway device **310** may communicate with the router **130** via a wired or wireless connection. Any number of devices in FIG. 3A, such as, for example, the router **130**, the gateway device **310**, laptop **314**, dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C**, among other devices, may be connected to the AC power supply **102**, perhaps via a hardwired connection or via electrical outlets **316** and **316A**, for example. Dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** may operate lighting load **104A** lighting load **104B**,

and/or lighting load 104C, respectively, as described previously herein. Occupancy sensor 180 may communicate with the router 130 and/or dimmer switches 110A, 110B, and/or 110C, perhaps to adjust the intensity of one or more of the dimmer switches 110A, 110B, and/or 110C based on a detected occupancy of the environment 300A. A user may activate one or more of the buttons (soft buttons or hard buttons (e.g. physical buttons or manual operators)) on a remote control device 184, which may communicate with the router 130 and/or dimmer switches 110A, 110B, and/or 110C to adjust the intensity of one or more of the dimmer switches 110A, 110B, and/or 110C. And a user may override the occupancy sensor's 180 control of the dimmer switches 110A, 110B, and/or 110C, for example.

The router 130 may establish a non-public (or private) IP address for the router 130 and may establish an IP connection and corresponding respective private IP addresses with the dimmer switch 110A, 110B, and/or 110C, the gateway device 310, and the laptops 312 and/or 314. The router 130 may coordinate one or more of the respective private IP addresses with one or more IP connections (e.g., multimedia or data streams) that are received via the router's 130 public IP address (e.g., from the server 304 and/or 306). The router 130 may coordinate one or more of the respective public IP addresses (e.g., of the server 304 and/or server 306) with one or more IP connections (e.g., multimedia or data streams) that are sent to the router's 130 private IP address (e.g., from the gateway device 310, laptop 312, and/or laptop 314). The router 130 may perform such coordination via a Network Address Table (NAT) (not shown), or the like, for example.

The wireless control device 120, the occupancy sensor 180, and/or the remote control device 184 may be operable to transmit and receive RF signals 106 including Internet Protocol packets directly to dimmer switches 110A, 110B, and/or 110C, or to dimmer switches 110A, 110B, and/or 110C via the wireless router 130 (and perhaps also via the gateway device 310). The router 130 (and perhaps the gateway device 310) may be operable to transmit one or more digital messages via RF signals 106 that may correspond to the RF signals 106 received from the wireless control device 120, the occupancy sensor 180, and/or the remote control device 184. The one or more digital messages may be transmitted according to a proprietary RF communication protocol (such as, for example, the Clear Connect™ protocol) to the dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C via RF signals 108. The dimmer switch 110A, dimmer switch 110B and/or dimmer switch 110C may include a wireless communication module (e.g. circuit) operable to receive digital messages according to the proprietary RF communication protocol via the RF signals 108. For example, the wireless control device 120, the occupancy sensor 106, the remote control device 184, the router 130, the laptop 312, and/or the laptop 314 may transmit the RF signals 106 directly to the dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C via a point-to-point communication, such as a Wi-Fi communication link, e.g., an 802.11 wireless local area network (LAN), or other direct wireless communication link, e.g., a Wi-MAX communication link or a Bluetooth® communication link.

In FIG. 3A, a communication dongle (not shown) could be connected to the wireless control device 120 that may allow for direct communication between the wireless control device 120 and the dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C using the proprietary RF communication protocol via RF signals 108. For example, the communication dongle could be plugged into a head-

phone jack on the wireless control device 120, or a USB port on 120. The occupancy sensor 180 and/or the remote control device 184 may communicate with the dimmer switches 110A, 110B, and/or 110C using the proprietary RF communication protocol via RF signals 108.

FIG. 3B is a diagram of an exemplary network environment 300B. In FIG. 3B, the router 130 may communicate with one or more servers 304, 306 via the Internet 308, perhaps as accessed through the "cloud." For example, router 130 may establish at least one Internet Protocol respective public IP addresses of server 304 and/or server 306). A gateway device 310 may communicate with the router 130 via a wired or wireless connection. Any number of devices in FIG. 3B, such as, for example, the router 130, the gateway device 310, laptop 314, dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C, among other devices, may be connected to the AC power supply 102, perhaps via a hardwired connection or via electrical outlets 316 and 316A, for example. Dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C may operate lighting load 104A lighting load 104B, and/or lighting load 104C as described previously herein. Occupancy sensor 180 may communicate with the router 130 and/or dimmer switches 110A, 110B, and/or 110C, perhaps to adjust the intensity of one or more of the dimmer switches 110A, 110B, and/or 110C based on a detected occupancy of the environment 300B. A user may activate one or more of the buttons (soft buttons or hard buttons (e.g. physical buttons or manual operators)) on the remote control device 184, which may communicate with the router 130 and/or dimmer switches 110A, 110B, and/or 110C to adjust the intensity of one or more of the dimmer switches 110A, 110B, and/or 110C. And a user may override the occupancy sensor's 180 control of the dimmer switches 110A, 110B, and/or 110C, for example.

The router 130 may establish a non-public (or private) IP address for the router 130 and may establish an IP connection and corresponding respective private IP addresses with the dimmer switches 110A, 110B, and/or 110C, the gateway device 310, and the laptop 312 and/or the laptop 314. The router 130 may coordinate one or more of the respective private IP addresses with one or more IP connections (e.g., multimedia or data streams) that are received via the router's 130 public IP address (e.g., from the server 304 and/or 306). The router 130 may coordinate one or more of the respective public IP addresses (e.g., of the server 304 and/or server 306) with one or more IP connections (e.g., multimedia or data streams) that are sent to the router's 130 private IP address (e.g., from the gateway device 310, laptop 312, and/or laptop 314). The router 130 may perform such coordination via a Network Address Table (NAT) (not shown), or the like, for example.

The wireless control device 120, the occupancy sensor 180, and/or the remote control device 184 may be operable to transmit and receive RF signals 106 including Internet Protocol packets directly to and from dimmer switches 110A, 110B, and/or 110C, or to and from the dimmer switches 110A, 110B, and/or 110C via the wireless router 130 (and perhaps also via the gateway device 310). The router 130 (and perhaps the gateway device 310) may be operable to transmit one or more digital messages via RF signals 106 that may correspond to the RF signals 106 received from the wireless control device 120, the occupancy sensor 180, and/or the remote control device 184. For example, the wireless control device 120, the occupancy sensor 106, the remote control device 184, the router 130, the laptop 312, and/or the laptop 314 may transmit the RF

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signals **106** directly to the dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** via a point-to-point communication, such as a Wi-Fi communication link, e.g., an 802.11 wireless local area network (LAN), or other direct wireless communication link, e.g., a Wi-MAX communication link or a Bluetooth® communication link.

The wireless control device **120**, the wireless router **130**, and the gateway device **310**, the occupancy sensor **180**, and/or the remote control device **184** may communicate with the laptop **314**, dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** via one or more devices that have a private IP address and are connected to the AC power source **102** via an Ethernet IP based protocol (e.g., the TCP/IP and/or “HomePlug” protocols) that may be carried via the conductors that deliver electrical energy from the AC power source **102** to the various devices (e.g., router **130**, gateway device **310**, dimmer switch **110A**, dimmer switch **110B**, dimmer switch **110C**, and/or laptop **314**). The gateway device **310**, the occupancy sensor **180**, the remote control device **184**, and the dimmer switches **110A**, **110B**, and/or **110C** (and perhaps the occupancy sensor **180** and the remote control device **184**) may also transmit, receive, and/or interpret energy pulses that may be used to convey signals and/or information via the conductors may deliver electrical energy from the AC power source **102** to the gateway device **310** and the dimmer switches **110A**, **110B**, and/or **110C** (and perhaps the occupancy sensor **180** and the remote control device **184**).

FIG. 3C is a diagram of an exemplary network environment **300C**. In FIG. 3C, the router **130** may communicate with one or more servers **304**, **306** via the Internet **308**, perhaps as accessed through the “cloud.” For example, router **130** may establish at least one Internet Protocol respective public IP addresses of server **304** and/or server **306**. A gateway device **310** may communicate with the router **130** via a wired or wireless connection. Any number of devices in FIG. 3C, such as, for example, the router **130**, the gateway device **310**, laptop **314**, dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C**, among other devices, may be connected to the AC power supply **102**, perhaps via a hardwired connection or via electrical outlets **316** and **316A**, for example. Dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** may operate lighting load **104A** lighting load **104B**, and/or lighting load **104C** as described previously herein. Occupancy sensor **180** may communicate with the router **130** and/or dimmer switches **110A**, **110B**, and/or **110C**, perhaps to adjust the intensity of one or more of the dimmer switches **110A**, **110B**, and/or **110C** based on a detected occupancy of the environment **300C**. A user may activate one or more of the buttons (soft buttons or hard buttons (e.g., physical buttons or manual operators)) on the remote control device **184**, which may communicate with the router **130** and/or dimmer switches **110A**, **110B**, and/or **110C** to adjust the intensity of one or more of the dimmer switches **110A**, **110B**, and/or **110C**. And a user may override the occupancy sensor’s **180** control of the dimmer switches **110A**, **110B**, and/or **110C** by activating one or more of the buttons of the remote control device **184**, for example.

The router **130** may establish a non-public (or private) IP address for the router **130** and may establish an IP connection and corresponding respective private IP addresses with the gateway device **310**, the laptop **312** and/or the laptop **314**. The router **130** may coordinate one or more of the respective private IP addresses with one or more IP connections (e.g., multimedia or data streams) that are received via the router’s **130** public IP address (e.g., from the server **304**

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and/or **306**). The router **130** may coordinate one or more of the respective public IP addresses (e.g., of the server **304** and/or server **306**) with one or more IP connections (e.g., multimedia or data streams) that are sent to the router’s **130** private IP address (e.g., from the gateway device **310**, laptop **312**, and/or laptop **314**). The router **130** may perform such coordination via a Network Address Table (NAT) (not shown), or the like, for example.

The wireless control device **120**, the occupancy sensor **180**, and/or the remote control device **184** may be operable to transmit and receive RF signals **106** including Internet Protocol packets directly to dimmer switches **110A**, **110B**, and/or **110C**, or to dimmer switches **110A**, **110B**, and/or **110C** via the gateway device **310** (and perhaps via the wireless router **130**). The gateway device **310** may be operable to transmit one or more digital messages via RF signals **106** that may correspond to the RF signals **106** received from the wireless control device **120**, the occupancy sensor **180**, and/or the remote control device **184** (perhaps via the router **130**). The one or more digital messages may be transmitted according to a proprietary RF communication protocol (such as, for example, the Clear Connect™ protocol) to the dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** via RF signals **108**. The dimmer switch **110A**, dimmer switch **110B** and/or dimmer switch **110C** may include a wireless communication module (e.g. circuit) operable to receive digital messages according to the proprietary RF communication protocol via the RF signals **108**. The gateway device **310** (and perhaps the router **130**) may communicate with the laptop **314**, dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** via an Ethernet based IP protocol (e.g., TCP/IP and/or “HomePlug” protocols) that may be carried via the conductors that deliver electrical energy from the AC power source **102** to the various devices such as the router **130**, the gateway device **310**, laptop **314**, dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C**, among other devices illustrated in FIG. 3C.

In FIG. 3C, a communication dongle (not shown) could be connected to the wireless control device **120** that may allow for direct communication between the wireless control device **120** and the dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** using the proprietary RF communication protocol via RF signals **108**. For example, the communication dongle could be plugged into a head-phone jack on the wireless control device **120**, or a USB port on **120**. The occupancy sensor **180** and/or the remote control device **184** may communicate with the dimmer switches **110A**, **110B**, and/or **110C** using the proprietary RF communication protocol via RF signals **108**.

The router **130** may further establish IP connections and corresponding respective private IP addresses with the occupancy sensor **180**, remote control device **184**, dimmer switch **110A**, **110B**, and/or **110C**. In such situations, the router **130** may coordinate one or more of the respective private IP addresses of the occupancy sensor **180**, remote control device **184**, dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** with one or more IP connections (e.g., multimedia or data streams) that are received via the router’s **130** public IP address (e.g., from the server **304** and/or **306**). The router **130** may coordinate one or more of the respective public IP addresses (e.g., of the server **304** and/or server **306**) with one or more IP connections (e.g., multimedia or data streams) that are sent to the router’s **130** private IP address (e.g., from the occupancy sensor **180**, remote control device **184**, dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C**).

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When dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C may be assigned private IP addresses, the wireless control device 120, the occupancy sensor 180, and/or the remote control device 184 (among other devices with private IP addresses) may transmit RF signals 106 including Internet Protocol packets to the dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C. For example, the wireless control device 120, the occupancy sensor 106, the remote control device 184, the router 130, the laptop 312, and/or the laptop 314 may transmit the RF signals 106 directly to the dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C via a point-to-point communication, such as a Wi-Fi communication link, e.g., an 802.11 wireless local area network (LAN), or other direct wireless communication link, e.g., a Wi-MAX communication link or a Bluetooth® communication link. The wireless control device 120, the occupancy sensor 180, and/or the remote control device 184 may communicate with the laptop 314, dimmer switch 110A, dimmer switch 110B, and/or dimmer switch 110C via one or more devices that have a private IP address and are connected to the AC power source 102 via an Ethernet IP based protocol (e.g., the TCP/IP and/or “HomePlug” protocols) that may be carried via the conductors that deliver electrical energy from the AC power source 102 to the various devices (e.g., router 130, gateway device 310, dimmer switch 110A, dimmer switch 110B, dimmer switch 110C, and/or laptop 314).

FIG. 4A is a simplified block diagram of a first example of a dimmer switch 400A (e.g., one of the dimmer switches 110A, 110B, 110C shown in FIG. 3A). The example dimmer switch 400A comprises a controllably conductive device 410 coupled in series electrical connection between the AC power source 102 and the lighting load 404 for control of the power delivered to the lighting load. The controllably conductive device 410 may comprise a relay or other switching device, or any suitable type of bidirectional semiconductor switch, such as, for example, a triac, a field-effect transistor (FET) in a rectifier bridge, or two FETs in anti-series connection. The controllably conductive device 410 includes a control input coupled to a drive circuit 412.

The dimmer switch 400A further comprises a control circuit, e.g., a controller 414, coupled to the drive circuit 412 for rendering the controllably conductive device 410 conductive or non-conductive to thus control the power delivered to the lighting load 404. The controller 414 may comprise a microcontroller, a programmable logic device (PLD), a microprocessor, an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or any suitable processing device or control circuit. A zero-crossing detector 415 determines the zero-crossings of the input AC waveform from the AC power supply 402. A zero-crossing may be the time at which the AC supply voltage transitions from positive to negative polarity, or from negative to positive polarity, at the beginning of each half-cycle. The controller 414 receives the zero-crossing information from the zero-crossing detector 415 and provides the control inputs to the drive circuit 412 to render the controllably conductive device 410 conductive and non-conductive at predetermined times relative to the zero-crossing points of the AC waveform.

The controller 414 receives inputs from mechanical switches 416 that are mounted on a printed circuit board (not shown) of the dimmer switch 400A, and are arranged to be actuated by buttons (e.g., the toggle actuator 114 and the intensity adjustment actuator 116). The controller 414 also controls light-emitting diodes 418, which are also mounted on the printed circuit board. The light emitting diodes 418

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may be arranged to illuminate visual indicators (e.g., the visual indicators 118) on a front surface of the dimmer switch 400A, for example, through a light pipe structure (not shown). The controller 414 is also coupled to a memory 420 for storage of unique identifiers (e.g., the MAC address and the IP address) of the dimmer switch 400A, the SSID and the SSID password of the wireless LAN, instructions for controlling the lighting load 404, programming instructions for communicating via a wireless communication link, or the like. The memory 420 may be implemented as an external integrated circuit (IC) or as an internal circuit of the controller 414. A power supply 422 generates a direct-current (DC) voltage V_{CC} for powering the controller 414, the memory 420, and other low-voltage circuitry of the dimmer switch 400A.

The dimmer switch 400A further includes a wireless communication module (e.g. circuit) 430 for transmitting and receiving wireless signals (e.g., the RF signals 106 and/or 108) to and from a wireless device (e.g., the wireless control device 120, the gateway device 310, and/or the wireless router 130). For example, the wireless communication module 430 may be configured to communicate via a Wi-Fi communication link, a Wi-MAX communication link, a Clear Connect™ communication link, and/or a Bluetooth® communication link. The wireless communication module 430 may also include one or more other radio protocol modules (e.g. radios) that may be operable to communicate via a number of other protocols including Wi-Fi and/or a proprietary RF protocol such as the Clear Connect™ protocol. The dimmer switch 400A may further include a second wireless communication module (e.g. circuit) 432 that may be configured to communicate via a Wi-Fi communication link, a Wi-MAX communication link, a Clear Connect™ communication link, and/or a Bluetooth® communication link. The wireless communication module 432 may also include one or more other radio protocol modules (e.g. radios) that may be operable to communicate via a number of other protocols including Wi-Fi and/or a proprietary RF protocol such as the Clear Connect™ protocol.

When the wireless communication modules 430 and/or 432 comprise a Wi-Fi module, the controller 414 is operable to control the lighting load 104 in response to received digital messages in Wi-Fi packets (i.e., Internet Protocol packets received via the Wi-Fi signals). If both of the wireless communication modules 430 and 432 comprise Wi-Fi modules, the modules may communicate using different frequency channels. The wireless communication module 430 and/or 432 may comprise one or more RF transceivers and one or more antennas. Examples of antennas for wall-mounted dimmer switches are described in greater detail in U.S. Pat. No. 5,736,965, issued Apr. 7, 1998, and U.S. Pat. No. 7,362,285, issued Apr. 22, 2008, both entitled COMPACT RADIO FREQUENCY TRANSMITTING AND RECEIVING ANTENNA AND CONTROL DEVICE EMPLOYING SAME, the entire disclosures of which are hereby incorporated by reference.

The dimmer switch 400A further comprises an optical module (e.g. circuit) 440, such as an optical signal receiving circuit for example. The optical module 440 may be optically coupled to an optical receiver (e.g., the optical receiver 119). The optical module 440 may be coupled to the optical receiver 119 on the front surface of the dimmer switch 400A, for example, through a light pipe (not shown), such that the optical module 440 may receive the optical signals from the wireless control device 120 via the light pipe. For example, the optical module 440 may comprise a photodiode (not

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shown) that is responsive to the optical signals transmitted by the wireless control device **120**. In addition, the photodiode of the optical module **440** may be controlled by the controller **414**, so as to transmit optical signals to the wireless control device **120** (as will be described in greater detail below), for example.

The wireless control device **120** may control the controllably conductive device **410** using the optical signals and/or the digital messages received via the RF signals **106** and/or RF signals **108**. For example, the controller **414** may determine the module from which the signals are received, e.g., from the wireless communication module **430** and/or **432** or the optical module **440**, and the controllably conductive device **410** may be controlled based on those signals. The controller **414** may also transmit messages to the wireless control device **120** via optical signals or digital messages transmitted via the RF signals **106** and/or RF signals **108**. For example, the controller **414** of the dimmer switch **400A** may be used to transmit digital messages to the wireless control device **120** via wireless communication. The digital messages may include alerts and/or feedback and status information regarding the lighting load **104**. The digital messages may also include error messages or indications as to whether the dimmer switch **400A** is able to communicate via a wireless communication link or RF signals **106** and/or RF signals **108**, for example.

FIG. **4B** is a simplified block diagram of a second example of a dimmer switch **400B** (e.g., one of the dimmer switches **110A**, **110B**, **110C** shown in FIG. **3B**). The example dimmer switch **400B** comprises a controllably conductive device **410**, a drive circuit **412**, a controller **414**, a zero-crossing detector **415**, mechanical switches **416**, light-emitting diodes **418**, a memory **420**, a power supply **422**, and an optical module **440**. The elements within these devices, the functions of these devices, and/or interactions of and among these devices may be the same or similar as described with respect to FIG. **4A**.

The dimmer switch **400B** further includes a wireless communication module (e.g. circuit) **430** for transmitting and receiving RF signals (e.g., the RF signals **106**) to and from a wireless device (e.g., the wireless control device **120**, the gateway device **310**, and/or the wireless router **130**). For example, the wireless communication module **430** may be configured to communicate via a Wi-Fi communication link, a Wi-MAX communication link, a Clear Connect™ communication link, and/or a Bluetooth® communication link. The wireless communication module **430** may also include one or more other radio protocol modules (e.g. radios) that may be operable to communicate via a number of other protocols including Wi-Fi and/or a proprietary RF protocol such as the Clear Connect™ protocol.

The dimmer switch **400B** may further include a power line interface (e.g. circuit) module **434** for transmitting and receiving signals carried on the conductors connected to the AC power source **102** via an Ethernet IP based protocol (e.g. TCP/IP, and/or a power line communication protocol such as the “HomePlug” protocol) where the conductors may deliver electrical energy from the AC power source **402** to the dimmer switch **400B**. The power line interface module **434** may also transmit, receive, and/or interpret energy pulses that may be used to convey signals and/or information via the conductors may deliver electrical energy from the AC power source **402** to the dimmer switch **400B**.

When the wireless communication module **430** comprises a Wi-Fi module, the controller **414** is operable to control the lighting load **404** in response to received digital messages in Wi-Fi packets (i.e., Internet Protocol packets received via

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the Wi-Fi signals). The wireless communication module **430** may comprise one or more RF transceivers and one or more antennas.

The wireless control device **120** may control the controllably conductive device **410** using the optical signals, the digital messages received via the RF signals **106**, and/or digital messages received via the Ethernet IP based power line protocol (e.g., TCP/IP and/or “HomePlug” protocols). For example, the controller **414** may determine the module from which the signals are received, e.g., from the wireless communication module **430**, the power line interface module **434**, or the optical module **440**, and the controllably conductive device **410** may be controlled based on those signals. The controller **414** may also transmit messages to the wireless control device **120** via optical signals or digital messages transmitted via the RF signals **106**, and/or digital messages transmitted via the Ethernet IP based power line protocol. For example, the controller **414** of the dimmer switch **400B** may be used to transmit digital messages to the wireless control device **120** via wireless communication. The digital messages may include alerts and/or feedback and status information regarding the lighting load **404**. The digital messages may also include error messages or indications as to whether the dimmer switch **400B** is able to communicate via a wireless communication link or RF signals **106**, for example.

FIG. **4C** is a simplified block diagram of a third example of a dimmer switch **400C** (e.g., one of the dimmer switches **110A**, **110B**, **110C** shown in FIG. **3C**). The example dimmer switch **400C** comprises a controllably conductive device **410**, a drive circuit **412**, a controller **414**, a zero-crossing detector **415**, mechanical switches **416**, light-emitting diodes **418**, a memory **420**, a power supply **422**, and an optical module **440**. The elements within these devices, the functions of these devices, and/or interactions of and among these devices may be the same or similar as described with respect to FIG. **4A**.

The dimmer switch **400C** further includes a wireless communication module (e.g. circuit) **430** for transmitting and receiving RF signals (e.g., the RF signals **106** and/or **108**) to and from a wireless device (e.g., the wireless control device **120**, the gateway device **310**, and/or the wireless router **130**). For example, the wireless communication module **430** may be configured to communicate via a Wi-Fi communication link, a Wi-MAX communication link, a Clear Connect™ communication link, and/or a Bluetooth® communication link. The wireless communication module **430** may also include one or more other radio protocol modules (e.g. radios) that may be operable to communicate via a number of other protocols including Wi-Fi and/or a proprietary RF protocol such as the Clear Connect™ protocol. The dimmer switch **400C** may further include a second wireless communication module (e.g. circuit) **432** that may be configured to communicate via a Wi-Fi communication link, a Wi-MAX communication link, a Clear Connect™ communication link, and/or a Bluetooth® communication link. The wireless communication module **432** may also include one or more other radio protocol modules (e.g. radios) that may be operable to communicate via a number of other protocols including Wi-Fi and/or a proprietary RF protocol such as the Clear Connect™ protocol. The dimmer switch **400C** may further include a power line interface module (e.g. circuit) **434** for transmitting and receiving signals carried on the conductors connected to the AC power source **402** via an Ethernet IP based protocol (e.g. TCP/IP, and/or a power line communication protocol such as the “HomePlug” protocol) where the conductors may deliver

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electrical energy from the AC power source **402** to the dimmer switch **400C**. The power line interface module **434** may also transmit, receive, and/or interpret energy pulses that may be used to convey signals and/or information via the conductors may deliver electrical energy from the AC power source **402** to the dimmer switch **400C**.

When the wireless communication modules **430** and/or **432** comprise a Wi-Fi module, the controller **414** is operable to control the lighting load **404** in response to received digital messages in Wi-Fi packets (i.e., Internet Protocol packets received via the Wi-Fi signals). The wireless communication module **430** and/or **432** may comprise one or more RF transceivers and one or more antennas.

The wireless control device **120** may control the controllably conductive device **410** using the optical signals and/or the digital messages received via the RF signals **106** and/or RF signals **108**. For example, the controller **414** may determine the module from which the signals are received, e.g., from the wireless communication module **430** and/or **432** or the optical module **440**, and the controllably conductive device **410** may be controlled based on those signals. The controller **414** may also transmit messages to the wireless control device **120** via optical signals or digital messages transmitted via the RF signals **106** and/or RF signals **108**. For example, the controller **414** of the dimmer switch **400C** may be used to transmit digital messages to the wireless control device **120** via wireless communication. The digital messages may include alerts and/or feedback and status information regarding the lighting load **404**. The digital messages may also include error messages or indications as to whether the dimmer switch **400C** is able to communicate via a wireless communication link or RF signals **106** and/or RF signals **108**, for example.

FIG. 5A a first simplified example block diagram of an input device, e.g., a remote control device **500A** (such as, the remote control device **184** of FIGS. 3A-3C). The example remote control device **500A** may include devices such as a controller **514**, a memory **520**, a wireless communication module **530**, and/or a wireless communication module **532**. One or more of the elements within these devices, one or more of the functions of these devices, and/or one or more of the interactions of and among these devices may be the same or similar as described with respect to FIG. 4A. The remote control device **500A** may also include a battery power supply **550** that may provide electrical power to the one or more devices included in the remote control device **500A**, such as the controller **514**.

The example remote control device **500A** may also include buttons **552**, visual indicators **556**, and/or a battery **550**. The controller **514** of the remote control device **500A** may be configured to receive commands input via the one or more buttons **552**. The one or more buttons **552** may include one or more soft buttons or one or more hard buttons (e.g., physical buttons or manual operators). For example, the controller **514** may interpret inputs via the one or more buttons **552** as user commands intended for one or more devices (e.g., a dimmer switch). Again by way of example, a user may contact one button of the one or more buttons **552** of remote control device **500A** to order the appropriate dimmer switch (e.g., the dimmer switch **110A**) to adjust the intensity of a lighting load (e.g., the lighting load **104A**) to 50%, among many other configurable adjustments. The controller **514** of the remote control device **500A** may interpret the signal from the one button of the one or more buttons **552** as a command to order the dimmer switch **110A** to perform the adjustment to 50%.

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The controller **514** may communicate the command to the dimmer switch **110A** via one or more wireless signals sent via wireless communication module **530** and/or **532** (e.g., in a manner that is the same or similar to the functions described with respect to communication modules **430** and/or **432** as described with regard to FIG. 4A). The controller **514** of the remote control device **500A** may be configured to control one or more visual indicators **556** to provide the user with one or more feedback or status indications (e.g., at least for a period of time). For example, one indicator of the one or more indicators **556** may indicate (e.g. for some period of time) that one or more buttons **552** may have been activated by a user (e.g. as interpreted by the controller **514**). Also by way of example, one indicator of the one or more indicators **556** may indicate (e.g., for a period of time) that the dimmer switch **110A** has received the command from the controller **414** to perform an adjustment (e.g. as input by the user) of the lighting load **104A**. Also by way of example, one indicator of the one or more indicators **556** may indicate that that battery **550** is at a low level of charge.

FIG. 5B a second simplified example block diagram of an input device, e.g., a remote control device **550B** (such as the remote control device **184** of FIGS. 3A-3C). The remote control device **500B** may include one or more of the same or similar functional blocks as those included and described with respect to the remote control device **500A** of FIG. 5A. The one or more of the elements within these functional blocks, one or more of the functions of these functional blocks, and/or one or more of the interactions of and among these functional blocks may be the same or similar as described with respect to FIG. 4A and FIG. 5A. The remote control device **500B** may operate with just the wireless communication module **530** in lieu of both the wireless communication modules **530** and **532**.

FIG. 5C is a first simplified example block diagram of a sensor device, e.g., an occupancy sensor **500C** (such as the occupancy sensor **180** of FIGS. 3A-3C). The occupancy sensor **500C** may include one or more of the same or similar functional blocks as those included and described with respect to the remote control device **500A** of FIG. 5A. The one or more of the elements within these functional blocks, one or more of the functions of these functional blocks, and/or one or more of the interactions of and among these functional blocks may be the same or similar as described with respect to FIG. 4A and FIG. 5A.

The occupancy sensor **500C** may also include at least one sensor circuit **554**. The at least one sensor circuit **554** may detect the presence (or lack thereof) of people in a given area of sensor effectiveness. The controller **514** of the occupancy sensor **500C** may be configured to receive a signal from the at least one sensor **554**, interpret the signal as indicating a presence or absence of people in the given area of sensor effectiveness (perhaps for a period of time), and/or send one or more commands to other devices based on the interpreted presence of people or lack thereof. For example, should the controller **514** of the occupancy sensor **500C** interpret the at least one sensor **554** to report the lack of presence in the given area of effectiveness (perhaps for some period of time, e.g., 60 seconds), the controller may send respective commands to wireless devices, e.g., to one or more of the dimmer switches **110A**, **110B**, and/or **110C** to lower the respective intensities of the lighting loads **104A**, **104B**, and/or **104C** (e.g., shutoff all the lights when all people have left the room). Also by way of example, should the controller **514** of the occupancy sensor **500C** interpret the at least one sensor **554** to report a transition from a lack of any presence to the presence of at least one person in the given area of

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effectiveness, the controller may send respective commands to one or more of the dimmer switches **110A**, **110B**, and/or **110C** to increase the respective intensities of the lighting loads **104A**, **104B**, and/or **104C** (e.g., turn at least some of the lights when at least one person enters the area of sensor effectiveness). The controller **514** of the occupancy sensor **500C** may communicate the command to the dimmer switch **110A**, dimmer switch **110B**, and/or dimmer switch **110C** via one or more wireless signals sent via wireless communication module **530** and/or **532** (e.g., in a manner that is the same or similar to the functions described with respect to communication modules **430** and/or **432** as described with regard to FIG. **4A**).

FIG. **5D** a second simplified example block diagram of a sensor device (such as the occupancy sensor **180** of FIGS. **3A-3C**). The occupancy sensor **500D** may include one or more of the same or similar functional blocks as those included and described with respect to the occupancy sensor **500C** of FIG. **5C**. The one or more of the elements within these functional blocks, one or more of the functions of these functional blocks, and/or one or more of the interactions of and among these functional blocks may be the same or similar as described with respect to FIG. **4A**, FIG. **5A**, and FIG. **5A**. The occupancy sensor **500D** may operate with just the wireless communication module **530** in lieu of both the wireless communication modules **530** and **532**.

FIG. **5E** is a simplified example block diagram of a contemplated combination input and sensor device **500E** which may be employed in the environments of FIGS. **3A-3C**. The combination input/sensor device **500E** may include one or more of the same or similar functional blocks as those included and described with respect to the remote control device **500A** and **500B** and the occupancy sensor **500C** and **500D** that may be employed in the environments of FIGS. **3A-3C**. The one or more of the elements within these functional blocks, one or more of the functions of these functional blocks, and/or one or more of the interactions of and among these functional blocks may be the same or similar as described with respect to FIG. **4A**, FIG. **5A**, and FIG. **5C**.

As described previously, any of the devices of the network environments **300A-300C** of FIGS. **3A-3C** (e.g. wireless control device **120**, dimmer switches **110A**, **110B**, and/or **110C**, router **130**, occupancy sensor **180**, remote control **184**, laptops **312** and/or **314**, among others shown and not shown) for a number of contemplate purposes, may include one or more radios. For example, any of the devices of the network environments **300A-300C** may include at least one radio that may be operable to transmit via multiple protocols (e.g., the Wi-Fi and/or the Clear Connect™ protocols) over multiple communication networks, wired and/or wireless, which may be operable to communicate with the respective protocols. Alternatively or additionally, any of the devices of the network environments **300A-300C** may include at least one radio that may be operable to transmit/receive via at least one protocol (e.g., the Wi-Fi protocol) and at least a second radio that may be operable to transmit/receive via at least another protocol (e.g., a proprietary RF protocol like the Clear Connect™ protocol) over multiple communication networks, wired and/or wireless, that may be operable to communicate with the respective protocols.

One or more, or any, of the devices of the network environments **300A-300C** may serve as a master gateway node (e.g., may be elected by the other devices to serve as the master gateway node). The master gateway node may serve as a Dynamic Host Configuration Protocol (DHCP) node (or function), for example. The master gateway node

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may provide one or more, or any, of the other devices of the network environments **300A-300C** with information that may enable the one or more other devices to connect to the Wi-Fi network (e.g., an IP based protocol). By way of example, and not limitation, the master gateway node may provide the one or more devices of the network environments **300A-300C** with a service set identifier (SSID), an SSID password, a wireless security password or key value such as a WEP password/key or a WPA password/key, and/or an IP address, and/or other credentials or access information to enable the respective devices to connect (or register) to the Wi-Fi protocol network (e.g., via the router **130**). Such Wi-Fi access information may be preconfigured on any of the respective devices of the network environments **300A-300C**.

The Wi-Fi access information may be provided to the one or more devices of the network environments **300A-300C** via a reliable broadcast-capable RF protocol, such as the previously described Clear Connect™ protocol, either approximately at a time that it may be useful for the one or more devices to join the Wi-Fi communication network, or at some time earlier. For example, the Wi-Fi access information (e.g., even if preconfigured) for the one or more devices may be updated by the master gateway node either periodically or under certain conditions. Also, the master gateway node may provide an indication (e.g., via the Clear Connect™ protocol) to the one or more devices of the network environments **300A-300C** that may invite the one or more devices to use the Wi-Fi protocol access information to communicate, at least temporarily (e.g., for a firmware upgrade), with one or more devices of the network environments **300A-300C** (e.g., the master gateway node or any other device of the network environments **300A-300C**). For example, perhaps after the invited node may have completed the function for which it was invited to join the Wi-Fi network (e.g., a firmware upgrade is fully communicated and/or completed), the master gateway node may signal (e.g., via the Wi-Fi and/or Clear Connect™ protocols) the invited node to discontinue Wi-Fi communication and/or to leave the Wi-Fi network. By requesting that the invited node discontinue Wi-Fi communication and/or to leave the Wi-Fi network, the burden on the router **130** and/or Wi-Fi communication may be minimized. Alternatively or additionally, the invited node may be configured to discontinue Wi-Fi communication and/or to leave the Wi-Fi network after the completion of the function for which it was invited to communicate via Wi-Fi and/or after the end of a timeout period (e.g., the invited node may leave the Wi-Fi network on its own determination and without being requested to leave the Wi-Fi network).

Alternatively or additionally, the one or more devices of the network environments **300A-300C** may use the Wi-Fi access information to communicate with one or more other devices of the network environments **300A-300C** at a time and/or under a condition determined by the one or more devices of the network environments **300A-300C** that may be in possession of Wi-Fi access information. For example, dimmer switch **110A** may use its respective Wi-Fi access information to join the Wi-Fi communication network to communicate data or information (e.g., to communication monitoring database information to one or other devices of the network environments **300A-300C**) via the Wi-Fi protocol, (e.g., perhaps because its monitoring database may have become full). After the dimmer switch **110A** communicates the data or information, the dimmer switch **110A** may discontinue communication via the Wi-Fi protocol until such time as the dimmer switch **110A** may be invited to (or

may decide itself to) communicate once again via the Wi-Fi protocol. In addition, when a button of an input device (e.g., the remote control device **184**) is actuated, the input device may use its respective Wi-Fi access information to join the Wi-Fi communication network to communicate information regarding an actuation of the button. Further, a sensor device (e.g., the occupancy sensor **180**) may use its respective Wi-Fi access information to join the Wi-Fi communication network to communicate information regarding an occupancy or vacancy condition. Alternatively, a sensor device, such as a daylight sensor, may periodically use its respective Wi-Fi access information to join the Wi-Fi communication network to communicate information regarding an ambient light intensity measured in a space.

The Wi-Fi protocol may be useful via which to communicate high bandwidth data (e.g. configuration data such as firmware upgrades and/or data for relatively sophisticated user interfaces, programming data, and/or database data management) among Wi-Fi capable (IP capable) devices. A reliable broadcast-capable RF protocol, such as the previously described Clear Connect™ protocol may be useful via which to communicate relatively low bandwidth data and/or relatively high performance signaling information (e.g. operational data such as operational commands, operational (runtime) error codes, programming error codes, and/or timing synchronization signals, among other relatively high performance data). It may be useful to allocate high bandwidth data signaling (e.g. firmware upgrades, user interface data, and/or database information transfer) more to Wi-Fi protocol communication so that reliable broadcast-capable RF protocol communication, such as via the Clear Connect™ protocol, may be allocated for the relatively high performance data signaling (e.g. time synchronization signaling).

For example, radios using the Wi-Fi protocol may communicate at a frequency of 2.4 GHz. This frequency may be considered part of the industrial, scientific, and medical (ISM) radio band-which may fairly crowded, may be widely available, and may be generally considered to be an unlicensed band. Radios may communicate using the Wi-Fi protocol at a range of 120 to 300 feet (with 802.11n, up to double these ranges may be possible), for example. Radios may communicate using the Wi-Fi protocol at a rate of up to 54 Mbits/s (802.11g) and/or 300 Mbit/s (802.11n), with an average data rate of approximately 22 Mbit/s, for example. Radios may communicate via Wi-Fi with an output power of approximately 20-100 mW (13-20 dBm).

For example, radios using the Clear Connect™ protocol may communicate at frequencies of 434 MHz and/or 868 MHz (perhaps based on regional factors). The 434 MHz and 868 MHz bands may be far less crowded than other bands and may be licensed, and may be subject to a relatively stringent set of regulations, including the United States' Federal Communications Commission (FCC) regulations that may limit transmit power and/or duty cycle, for example. Radios may communicate using the Clear Connect™ protocol at a range of 30 to 60 feet indoor and/or 300 feet open air (perhaps extendable via repeaters), for example. Radios may communicate using the Clear Connect™ protocol at a rate of up to 62.5 Kbit/s, for example. Radios may communicate via the Clear Connect™ protocol with an output power of approximately 4 mW (5 dBm).

FIG. 6 is a diagram that illustrates wireless communication network access activity for one or more contemplated devices and techniques. At **6002**, one or more devices, such as the dimmer switch **110A**, dimmer switch **110B**, dimmer switch **110C**, the occupancy sensor **180**, and/or the remote

control device **184**, and other devices from environments **300A-300C** (both shown and not shown), may not be communicating via a wireless communication network (e.g., a wireless local area network (LAN)) in which communication may be conducted via a particular wireless protocol, for example the Wi-Fi protocol. By not communicating on the Wi-Fi communication network, the respective device or devices may not place a burden on the Wi-Fi communication network and/or the router **130**. While the respective device or devices may not be communicating on the Wi-Fi communication network, the device or devices may or may not be operating on another wireless communication network (e.g. a Clear Connect™ protocol network) and/or a wired communication network (e.g., a wired TCP/IP network or a power line protocol network). At **6004**, one or more of the devices may, perhaps to execute a particular task, receive a trigger from another device to access the Wi-Fi communication network. Alternatively or additionally, the device may determine an internal trigger condition to access the Wi-Fi communication network.

At **6006**, the device or devices may access the Wi-Fi communication network, perhaps with access information provided by another device or preconfigured on the accessing device or devices. While on the Wi-Fi communication network the device or devices may execute the tasks or tasks that may have been part of the trigger for the device or devices to access the Wi-Fi communication network. At **6008**, perhaps either after completing the task, after some predetermined time, and/or after receiving a disconnect signal from another device, the device or devices may leave the Wi-Fi communication network.

Referring to FIG. 7 and in view of FIGS. 3A-3C, a contemplated technique **7000** may start at **7002** and may include, at **7004**, sending Wi-Fi connection information (e.g., from the master gateway node) to a targeted device of a network environment (e.g., dimmer switch **110A** or other load control device) via a proprietary RF protocol (e.g., the Clear Connect™ protocol). At **7006**, another node in the network environment may decide to communicate a firmware upgrade to the targeted device. At **7008**, the targeted node may be signaled (e.g. via the master gateway node) to use the Wi-Fi connection information to establish a Wi-Fi communication connection (e.g., with the router **130**) via a proprietary RF protocol (e.g., Clear Connect™ protocol).

At **7010**, the targeted device may use the Wi-Fi connection information to establish itself on the Wi-Fi network and commence Wi-Fi communication (e.g., via the IP address provided by the master gateway node). At **7012**, send the firmware upgrade data to the targeted device via the Wi-Fi communication network (e.g., the master gateway device and/or via the IP address of the targeted device). At **7014**, upon the completion of the firmware upgrade data transfer and/or a successful firmware upgrade, the targeted device may receive a request (e.g., via the master gateway node) to terminate Wi-Fi communication and/or leave the Wi-Fi network. Alternatively or additionally, at **7014**, upon the completion of the firmware upgrade data transfer and/or a successful firmware upgrade, the targeted device may determine a condition and/or period of time to terminate Wi-Fi communication and/or leave the Wi-Fi network. At **7016**, the targeted device may terminate the Wi-Fi connection and/or leave the Wi-Fi network upon the condition being satisfied or the request being received. At **7018**, the technique may end and may resume at **7002** as often as required to accommodate user configured load control functions for the network environments **300A-300C**.

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While the present application has been described with reference to the dimmer switches **110**, and the wireless control devices **120**, the concepts of the contemplated devices and techniques could be applied to any control devices that are operable to communicate with each other, such as, for example, dimming ballasts for driving gas-discharge lamps; light-emitting diode (LED) drivers for driving LED light sources; screw-in luminaires including integral dimmer circuits and incandescent or halogen lamps; screw-in luminaires including integral ballast circuits and compact fluorescent lamps; screw-in luminaires including integral LED drivers and LED light sources; electronic switches, controllable circuit breakers, or other switching devices for turning appliances on and off; plug-in load control devices, controllable electrical receptacles, or controllable power strips for each controlling one or more plug-in loads; motor control units for controlling motor loads, such as ceiling fans or exhaust fans; drive units for controlling motorized window treatments or projection screens; motorized interior or exterior shutters; thermostats for a heating and/or cooling systems; temperature control devices for controlling setpoint temperatures of HVAC systems; air conditioners; compressors; electric baseboard heater controllers; controllable dampers; humidity control units; dehumidifiers; water heaters; pool pumps; televisions; computer monitors; audio systems or amplifiers; generators; electric chargers, such as electric vehicle chargers; an alternative energy controllers; occupancy sensors, vacancy sensors, daylight sensors, temperature sensors, humidity sensors, security sensors, proximity sensors, keypads, battery-powered remote controls, key fobs, cell phones, smart phones, tablets, personal digital assistants, personal computers, timeclocks, audio-visual controls, safety devices, and central control transmitters.

Additionally, the techniques described herein may be implemented as a set of computer-executable instructions stored on a computer-readable medium, such as a random-access or read-only memory for example. Such computer-executable instructions may be executed by a processor or microcontroller, such as a microprocessor, within the dimmer switch **110** or the wireless control device **120**, for example.

What is claimed is:

1. An electrical load control device, comprising:
 - first communications interface circuitry communicatively coupleable to a first network using a first communication protocol;
 - second communications interface circuitry communicatively coupleable to a second network using a second communication protocol different from the first communication protocol;
 - memory circuitry to store configuration information;
 - control circuitry communicatively coupled to the first communications interface circuitry and to the second communications interface circuitry, the control circuitry to:
 - receive a first instruction to communicatively couple to the second network;
 - communicatively couple, via the second communications interface circuitry, to the second network;
 - receive, a second instruction to update one or more configuration parameters;
 - receive, via the second communications interface circuitry, one or more updated configuration parameters;
 - update the stored configuration information with the one or more updated configuration parameters;

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terminate the connection with the second network responsive to at least one of:

receipt of an instruction to terminate the connection to the second network; or

fulfillment of at least one condition precedent to terminate the connection to the second network; and

communicatively couple, via the first network circuitry, to the first network.

2. The electrical load control device of claim **1** wherein to receive the first instruction to communicatively couple to the second network, the control circuitry to further:

receive, via the first communications interface circuitry, the first instruction to communicatively couple to the second network, the first instruction communicated by a peer electrical load control device communicatively coupled to the first network.

3. The electrical load control device of claim **1** wherein to receive the first instruction to communicatively couple to the second network, the control circuitry to further:

receive the first instruction to communicatively couple to the second network from a portable electronic device communicatively coupled to the electrical load control device.

4. The electrical load control device of claim **3**:

wherein the electrical load control device further comprises optical communications interface circuitry; and wherein to receive the first instruction to communicatively couple to the second network from the portable electronic device, the control circuitry to further:

receive, from the portable electronic device via the optical communications circuitry, an optical communication that includes the first instruction to communicatively couple to the second network.

5. The electrical load control device of claim **3**:

wherein the electrical load control device further comprises third communications interface circuitry; and wherein to receive the first instruction to communicatively couple to the second network from the portable electronic device, the control circuitry to further:

receive, from the portable electronic device via the third communications interface circuitry, at least one of: a Bluetooth® message or a Near-Field Communication (NFC) message that includes the first instruction to communicatively couple to the second network.

6. The electrical load control device of claim **1** wherein to receive the first instruction to communicatively couple to the second network, the control circuitry to further:

receive, via the first communications interface circuitry, the first instruction to communicatively couple to the second network, the first instruction communicated by an electrical load control system controller communicatively coupled to the first network.

7. The electrical load control device of claim **1** wherein to receive the first instruction to communicatively couple to the second network, the control circuitry to further:

receive, via the first communications interface circuitry using a proprietary communication protocol, the first instruction to communicatively couple to the second network.

8. The electrical load control device of claim **1** wherein to communicatively couple to the second network via the second communications interface circuitry, the control circuitry to further:

communicatively couple to the second network via the second communications interface circuitry using a TCP/IP communication protocol.

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9. The electrical load control device of claim 1 wherein to receive the first instruction to communicatively couple to the second network, the control circuitry to further:

receive the first instruction to communicatively couple to the second network via actuation of an actuator disposed on the surface of the electrical load control device.

10. The electrical load control device of claim 1 wherein to receive the first instruction to communicatively couple to the second network, the control circuitry to further:

terminate an established communicative coupling with the first network prior to communicatively coupling to the second network.

11. A method of updating one or more configuration parameters in an electrical load control device, the method comprising:

receiving, by control circuitry in an electrical load control device, a first instruction to communicatively couple to a second network;

communicatively coupling, by the control circuitry, to the second network;

receiving, by the control circuitry via the second network, a second instruction to update one or more configuration parameters;

receiving, by the control circuitry via the second network, the one or more updated configuration parameters;

updating, by the control circuitry, configuration information stored in memory circuitry communicatively coupled to the control circuitry, with the one or more updated configuration parameters;

terminating, by the control circuitry, the connection with the second network responsive to at least one of:

receipt of an instruction to terminate the connection to the second network; or

fulfillment of at least one condition precedent to terminate the connection to the second network; and

communicatively coupling, by the control circuitry, to the first network.

12. The method of claim 11 wherein receiving the first instruction to communicatively couple to the second network further comprises:

receiving, by the control circuitry, the first instruction to communicatively couple to the second network from a peer electrical load control device communicatively coupled to the first network.

13. The method of claim 11 wherein receiving the first instruction to communicatively couple to the second network further comprises:

receiving, by the control circuitry, the first instruction to communicatively couple to the second network from a portable electronic device communicatively coupled to the electrical load control device.

14. The method of claim 13 wherein receiving the first instruction to communicatively couple to the second network from the portable electronic device, further comprises:

receiving, from optical communications circuitry communicatively coupled to the control circuitry, an optical communication that includes the first instruction to communicatively couple to the second network.

15. The method of claim 13 wherein receiving the first instruction to communicatively couple to the second network from the portable electronic device, further comprises:

receiving, from third communications interface circuitry communicatively coupled to the control circuitry, at least one of: a Bluetooth® message or a Near-Field

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Communication (NFC) message that includes the first instruction to communicatively couple to the second network.

16. The method of claim 11 wherein receiving the first instruction to communicatively couple to the second network further comprises:

receiving, by the control circuitry, the first instruction to communicatively couple to the second network, the first instruction communicated by an electrical load control system controller communicatively coupled to the first network.

17. The method of claim 11 wherein receiving the first instruction to communicatively couple to the second network further comprises:

receiving, by the control circuitry, the first instruction to communicatively couple to the second network using a proprietary communication protocol.

18. The method of claim 11 wherein communicatively coupling to the second network further comprises:

communicatively coupling, by the control circuitry, to the second network using a TCP/IP communication protocol.

19. The method of claim 11 wherein receiving the first instruction to communicatively couple to the second network further comprises:

receiving, by the control circuitry, the first instruction to communicatively couple to the second network via manual actuation of an actuator operatively coupled to the electrical load control device.

20. The method of claim 11, further comprising:

terminating an established communicative coupling with the first network responsive to receipt of the first instruction to communicatively couple to the second network.

21. A non-transitory, machine-readable, storage device that includes instructions that, when executed by control circuitry operatively coupled to an electrical load control device, cause the control circuitry to:

receive a first instruction to communicatively couple to a second network;

communicatively couple to the second network responsive to receipt of the first instruction;

receive a second instruction to update one or more configuration parameters via the second network;

receive the one or more updated configuration parameters via the second network;

update configuration information stored in memory circuitry communicatively coupled to the control circuitry, with the one or more updated configuration parameters;

terminate the connection with the second network responsive to at least one of:

receipt of an instruction to terminate the connection to the second network; or

fulfillment of at least one condition precedent to terminate the connection to the second network; and

communicatively couple to the first network.

22. The non-transitory, machine-readable, storage device of claim 21 wherein the instructions that cause the control circuitry to receive the first instruction to communicatively couple to the second network further cause the control circuitry to:

receive, via a first network, the first instruction to communicatively couple to the second network from a peer electrical load control device communicatively coupled to the first network.

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23. The non-transitory, machine-readable, storage device of claim 21 wherein the instructions that cause the control circuitry to receive the first instruction to communicatively couple to the second network further cause the control circuitry to:

receiving the first instruction to communicatively couple to the second network from a portable electronic device communicatively coupled to the electrical load control device.

24. The non-transitory, machine-readable, storage device of claim 23 wherein the instructions that cause the control circuitry to receive the first instruction to communicatively couple to the second network from the portable electronic device, further cause the control circuitry to:

receive, via communicatively coupled optical communications circuitry, an optical communication that includes the first instruction to communicatively couple to the second network.

25. The non-transitory, machine-readable, storage device of claim 23 wherein the instructions that cause the control circuitry to receive the first instruction to communicatively couple to the second network from the portable electronic device, further cause the control circuitry to:

receive at least one of: a Bluetooth™ message or a Near-Field Communication (NFC) message that includes the first instruction to communicatively couple to the second network.

26. The non-transitory, machine-readable, storage device of claim 21 wherein the instructions that cause the control circuitry to receive the first instruction to communicatively couple to the second network further cause the control circuitry to:

receive the first instruction to communicatively couple to the second network, the first instruction communicated

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by an electrical load control system controller communicatively coupled to the first network.

27. The non-transitory, machine-readable, storage device of claim 21 wherein the instructions that cause the control circuitry to receive the first instruction to communicatively couple to the second network further cause the control circuitry to:

receive the first instruction to communicatively couple to the second network using a proprietary communication protocol.

28. The method of claim 11 The non-transitory, machine-readable, storage device of claim 21 wherein the instructions that cause the control circuitry to communicatively couple to the second network further cause the control circuitry to:

communicatively couple to the second network using a TCP/IP communication protocol.

29. The non-transitory, machine-readable, storage device of claim 21 wherein the instructions that cause the control circuitry to receive the first instruction to communicatively couple to the second network further cause the control circuitry to:

receive the first instruction to communicatively couple to the second network via manual actuation of an actuator operatively coupled to the electrical load control device.

30. The non-transitory, machine-readable, storage device of claim 21 wherein the instructions further cause the control circuitry to:

terminate an established communicative coupling with the first network responsive to receipt of the first instruction to communicatively couple to the second network.

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